

MATH 110: APPLIED COLLEGE MATHEMATICS



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Math 110: Applied College Mathematics for Western Oregon University

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About This Book

Welcome to Math 110! Math 110 is a class designed specifically for Western Oregon University (WOU) students. In this class, you will gain mathematical confidence and skills that you can use for the rest of your life. Unlike a traditional mathematics course, Math 110 blends topics and techniques from a variety of mathematical subjects to answer a single question: how can mathematics be used to make decisions in common, real-world situations that you'll encounter in your life and career?

Each chapter has several sections. In each section, you will find learning objectives for that section, an exposition of that section's topic, and several worked-out examples. At the end of each section, there are homework exercises. In addition to this book, there are also [videos](#) to assist with student learning. Your instructor will provide specific instructions and guidance about how to work through these materials in a way unique to the timing and delivery of your class section.

Even if you're not confident in your ability to do mathematics, if you take the time to study these notes and related materials, you will begin to see the power and utility of mathematics in your life. I invite you to keep an open mind, work hard, and see what mathematical thinking can do for you!

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Since then, the book has undergone numerous content and formatting changes. Many of these were suggested by the following scholars:

- [Dr. Sarah Hagen](#), Mathematics Teacher, Davidson Academy
- [Dr. Charles R.B. Wright](#), Professor Emeritus, University of Oregon
- [Kendall Rosales](#), Mathematics Instructor and Math Center Director, Western Oregon University

Additionally, many improvements have been suggested by students in Math 110 at Western Oregon University. The author also wishes to thank her parents for being the primary editors of the first edition of the book.

Do you have a suggestion for how to improve this book? Feel free to email your ideas to [Dr. Leanne Merrill](#) at merrill@wou.edu.

CHAPTER OVERVIEW

1: Numbers and Relationships

Learn about our number system and the various ways in which we write and compare numbers.

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1.1: Our Number System

When we are children, we learn to count objects using ordinary numbers like 1, 2, 3, 4, so we can describe quantities in the world around us. As we get older, we learn larger and larger numbers, so we can make sense of statements like "the population of the world is about 8 billion people."

We also learn about other types of numbers, such as 0, negative numbers, and non-whole numbers, which help us describe even more precise quantities, such as "I walked 2.3 miles today." In this section, we'll briefly review the meaning of these numbers and the multiple ways to express them.



In this section, you will learn to:

- Accurately describe the meaning of the base 10 number system, including both whole numbers and non-whole numbers
- Correctly perform base 10 addition with any numbers and explain why it works
- Round numbers correctly to any place

1.1.1: The Base 10 Number System

You're already an expert at addition — you've been adding numbers for at least a decade, if not much longer. But when's the last time you really thought about how addition works?

Many of us routinely rely on technology such as smartphones or calculators to perform basic addition. It's faster and less error-prone than using mental math or pencil and paper, particularly when dealing with large numbers. The purpose of this section is not to discourage use of technology, but rather to peek behind the curtain of how humans, the creators of technology, designed the number system we all use every day.

Below, you'll encounter the first example in this book. Take a few minutes to think about the answer to the question in the example before viewing the solution. This example, as well as the many others throughout this book, will be most impactful to your learning if you attempt to solve them before looking ahead at the solution. In fact, [neuroscience has shown](#) that the more you struggle initially, the more likely you'll be to remember the reasoning behind the solution, which in turn leads to durable and applicable mathematical skill.



Example 1.1.1

Your seven-year-old cousin is learning to add numbers in school, and she asks you for help. She gives you the following problem:

$$\begin{array}{r} 374 \\ + 258 \\ \hline \end{array} \quad (1.1.1)$$

How would you explain the solution to her?

Solution

The most common method used to solve this problem is explained below. This might not be how you learned how to do this, but for illustration, here it is:

- Add the two numbers in the rightmost column;
- If they add up to more than 10, put the ones digit of the sum in that column, and "carry" the 1 to the column to the left;
- Repeat this procedure in the next column to the left, remembering to add in any numbers that had been carried;
- Keep going from right to left until you finish.

In terms of what you'd tell your cousin, you might say: "First, add together the 4 and the 8 in the rightmost column. That answer is 12, so you write down the 2 underneath, and then carry the 1." You could write it like this:

$$\begin{array}{r} 374 \\ + 258 \\ \hline \end{array} \quad \begin{array}{r} 1 \\ 12 \\ 2 \end{array} \quad (1.1.2)$$

$$\begin{array}{r} + \quad 2 \quad 5 \quad 8 \\ \hline 2 \end{array} \quad (1.1.2)$$

From there, you could explain: "Next, add together all of the numbers in the next column. That is $1 + 7 + 5$, which is 13. Just like before, write the 3 down below, and then carry the 1." You could write it like this:

$$\begin{array}{r} \quad 3 \quad \overset{1}{7} \quad 4 \\ + \quad 2 \quad 5 \quad 8 \\ \hline \quad 3 \quad 2 \end{array} \quad (1.1.3)$$

And then you could say: "Since you're out of columns, just add up the last numbers in that column, and put the answer underneath. That would give you $1 + 3 + 2 = 6$. So the answer is 632." Writing that would look like this:

$$\begin{array}{r} \quad 3 \quad \overset{1}{7} \quad 4 \\ + \quad 2 \quad 5 \quad 8 \\ \hline \quad 6 \quad 3 \quad 2 \end{array} \quad (1.1.4)$$

If you said that, your cousin would know exactly how to solve this sort of problem, and with practice she could likely solve many similar problems. It's a great explanation of *how* the process works.

But your cousin wants to know *why* the process works. She asks, "why do you carry the 1?"

Definition: Base 10 System

We use the **base 10** or **decimal** number system. In this system, each digit is a single whole number between 0 and 9 that represents a particular power of 10, with the power increasing from right to left. We typically refer to these powers of ten as **place values**.

What does this mean when referring to a particular number? For example, consider the number 4027, which would be read as "four thousand and twenty-seven." If we break this down into its place values, we have that 4 is in the thousands place, 0 is in the hundreds place, 2 is in the tens place, and 7 is in the ones place. Therefore, we could choose to rewrite this number as follows:

$$4027 = (4 \times 1000) + (0 \times 100) + (2 \times 10) + (7 \times 1) \quad (1.1.5)$$

Writing numbers in this way is more time-consuming than writing them the normal way. However, it allows us to see a pattern that emerges. In order to see that pattern more fully, recall the following definition:

Definition: Exponent

An **exponent** represents repeated multiplication a certain number of times. Exponents are written as superscripts. For example,

$$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32 \quad (1.1.6)$$

Notice that above, there are 5 copies of the number 2 being multiplied together. In general,

$$a^b = a \times a \times \cdots \times a \quad (b \text{ times}) \quad (1.1.7)$$

A nonzero number to the 0 power is defined to equal 1; for example $2^0 = 1$.

We introduce this definition because it allows us to describe the pattern that occurs in base 10 numbers. Observe — or better yet, check with your phone or a calculator — that applying larger and larger exponents to 10 simply increases the number of 0's following the 1. For example, computing 10^3 gives us

$$10^3 = 10 \times 10 \times 10 = 1000 \quad (1.1.8)$$

which has 3 zeros following the 1. This pattern occurs in general. The following table shows the first several occurrences of this pattern:

Exponent	Exponential Form	Ordinary Form
0	10^0	1
1	10^1	10
2	10^2	100
3	10^3	1000
4	10^4	10000

Note

In the United States, commas are used to separate groups of three digits in large numbers, starting from the ones place. That is, the number 10000 could be written 10,000 and both would be said as "ten thousand." In this text, we will typically omit commas unless it is helpful for comprehension.

Let's take this pattern and use it to further inspect the number 4027. We have

$$\begin{aligned} 4027 &= (4 \times 1000) + (0 \times 100) + (2 \times 10) + (7 \times 1) \\ 4027 &= (4 \times 10^3) + (0 \times 10^2) + (2 \times 10^1) + (7 \times 10^0) \end{aligned} \quad (1.1.9)$$

This is why our number system is called *base 10*. Each successive digit, going from right to left, corresponds to a power of 10. When we write a number as we did above on the right side of the equations, we call it the *base 10 expansion* of that number. The numbers 4, 0, 2 and 7 are called the *digits* of the number.

✓ Example 1.1.2

Why do we use 10 as the base for our number system?

Solution

The answer lies at the ends of your arms! Most humans have 10 fingers, so it's very convenient to count to 10 and then start over again. If we had 8 fingers, our number system would likely be very different, and the numeral 9 might not even exist! This is also why our fingers are sometimes called *digits* themselves.

Other base systems are used in particular contexts. For example, *binary numbers* are written in base 2, in which each successive place corresponds to another power of 2. This is the number system used in computer and electronics programming, though it is hidden from most device users. We also see base 12 and base 60 showing up in our computations of time: when counting seconds and minutes, we count to 59 and then start over again at 0.

Nevertheless, base 10 is the most common way to express numbers for the vast majority of people in our world today.

Now that we better understand the base 10 system, we can return to your cousin's question: why do we carry the 1 when we add? The answer is quite simple: because of our number system, we can only write a single digit in a given place. Remember your cousin's addition problem:

$$\begin{array}{r} 3 \ 7 \ 4 \\ + \ 2 \ 5 \ 8 \\ \hline \end{array} \quad (1.1.10)$$

In adding the $4 + 8$, we get 12, but we can't put a 12 in the ones place, because 12 consists of two digits. Therefore, we have to remember what the number 12 actually *means*. We can rewrite 12 using the base 10 expansion to uncover that meaning:

$$12 = (1 \times 10^1) + (2 \times 10^0) \quad (1.1.11)$$

We can write the 2 in the ones place, since it's only one digit. However, the additional 1 in the tens place has to be added to the other values in the tens place, which in this case are 7 and 5. Since the sum of the digits in a single column exceed 9, the excess has to be "carried over" to the next column, which represents the next larger power of 10.

While these observations may seem very basic, these insights on how we represent numbers can have a big impact on how comfortable we feel working with more complicated types of numbers and making judgments about numerical questions.

1.1.2: Decimals

Decimal numbers are written using the same logic and patterns that whole numbers do. While technically anything written in base 10 is a "decimal number," people often use the word "decimal" to describe numbers that are not whole numbers, such as 98.7 or 510.106. Non-whole numbers exist between whole numbers, and the section of the number after the decimal point describes precisely where the number lies on a number line.

Let's investigate a number between 0 and 1 first: the number 0.439. The 0 to the left of the decimal point tells us that this number is between 0 and 1, and the part after the decimal point describes its value. The place values of this number work just like they do for whole numbers, except that instead of counting in groups of 10, 100, 1000, and so on, they stand for the *reciprocals* of those values. We use negative exponents to denote these reciprocals since it is shorter to write.

Definition: Negative Exponents

A **negative exponent** on a number indicates that you should find the reciprocal of that number. That is,

$$2^{-5} = \frac{1}{2^5} = \frac{1}{32} \quad (1.1.12)$$

In general,

$$a^{-b} = \frac{1}{a \times a \times \cdots \times a} \quad (b \text{ times}) \quad (1.1.13)$$

Let's use this definition in context to help us understand how decimals can be expressed using base 10 notation.

✓ Example 1.1.3

Rewrite 10^{-1} , 10^{-2} , and 10^{-3} without any exponents, and convert them to decimal form.

Solution

Using the definition of negative exponents along with the original definition of exponents, we see that

$$10^{-1} = \frac{1}{10^1} = \frac{1}{10} \quad (1.1.14)$$

$$10^{-2} = \frac{1}{10^2} = \frac{1}{100} \quad (1.1.15)$$

$$10^{-3} = \frac{1}{10^3} = \frac{1}{1000} \quad (1.1.16)$$

These numbers are valid fractions, but it's often easier to work with decimals in calculations. Each of these fractions corresponds to a decimal value in the following way. The pronunciation is included as well.

$$\frac{1}{10} = 0.1 = \text{one tenth} \quad (1.1.17)$$

$$\frac{1}{100} = 0.01 = \text{one hundredth} \quad (1.1.18)$$

$$\frac{1}{1000} = 0.001 = \text{one thousandth} \quad (1.1.19)$$

Note that the reciprocal of a number is named similarly to the number itself, with a "th" afterwards.

Of course, we could keep going with this pattern to ten thousandths, hundred thousandths, and so on.

These negative powers of 10 form the basis for how we write decimals, just as the positive powers of 10 give us a way to write any whole number. A final example is given next. Remember, you'll learn better if you try it first before looking at the answer!

✓ Example 1.1.4

Write 0.439 in expanded form, using negative powers of 10.

Solution

Each successive place in this number corresponds to a negative power of 10 as we saw in the previous example. The 4 following the decimal point is in the 0.1 place, also known as the $\frac{1}{10}$ place or the 10^{-1} place. Next, the 3 is in the 0.01 place, also known as the $\frac{1}{100}$ place or the 10^{-2} place. And last, the 9 is in the 0.001 place, also known as the $\frac{1}{1000}$ place or the 10^{-3} place.

We can therefore expand this number in the following way:

$$0.439 = 4 \times 0.1 + 3 \times 0.01 + 9 \times 0.001 = 4 \times 10^{-1} + 3 \times 10^{-2} + 9 \times 10^{-3} \quad (1.1.20)$$

Note the similarities and differences with a base 10 expansion of a whole number.

Now that we've got some tools to describe and compare whole numbers and decimals, we'll introduce one last important concept for the chapter.

1.1.3: Rounding

Rounding is a skill you've likely seen before, but may need to practice. Let's get started with two examples.

✓ Example 1.1.5

The population of Bluffington is 57,489 people. Express this fact as a sentence in which the population is rounded to the nearest thousand.

Solution

To round the number 57,489 to the nearest thousand, the first step is to locate the thousands place. After rounding, every number after the thousands place will be a 0.

In this case, the 7 is in the thousands place, because it stands for 7×1000 in the base 10 expansion of 57,489. Since the number in the thousands place is a 7, it tells us that the number 57,489 is between 57,000 and 58,000 and will be rounded to one of those two values.

The question is: which of 57,000 and 58,000 is 57,489 closer to? The answer is 57,000 because the number 57,489 is less than 57,500 which is the halfway point between 57,000 and 58,000.

So, as a sentence, we could write, "The population of Bluffington is approximately 57,000 people."

Try the next exercise on your own before looking at the answer.

✓ Example 1.1.6

In performing a calculation, you need to enter $45 \div 40$ into your calculator. This gives you the answer 1.125. However, the question you are answering with this calculation asks for an amount of money, in dollars and cents. Round the number 1.125 to the correct decimal place and value so it is expressed as a number of dollars and cents.

Solution

To answer this question, you'll need to know that money amounts in the United States are typically expressed as a number of dollars followed by a number of cents, where a dollar is composed of one hundred cents. That means that each cent is *one hundredth* of a dollar, so we will need to round to the hundredths place.

The hundredths place refers to the second number after a decimal point in a decimal number. In the number 1.125, the current value of the hundredths place is a 2. This tells us the number 1.125 is between 1.12 and 1.13.

However, we still need to ask what 1.125 is closest to: either 1.12 or 1.13. In reality, it's exactly halfway between these two values, but we adopt (as do most textbooks) the habit of "rounding up" from the halfway point. Therefore, the number 1.125 rounded to the nearest hundredth is 1.13. As a dollar amount, you would write \$1.13.

These two examples illustrate the general principles of rounding. The process in general is summed up in the Procedure box below.

Rounding Numbers

To round a number to a given place value, follow these steps:

- Identify the the original digit in the place value you're rounding to.
- Look at the digit to the *right* of the original digit.
 - If the digit to the right is 4 or less, keep the original digit in the place to which you are rounding.
 - If the digit to the right is 5 or more, increase the original digit by 1.
- All digits to the right of the original digit either become 0 (if you are rounding to a whole number) or go away (if you are rounding to a decimal value).

One more example will help to clarify. Consider the number 34.5718 We label the place values of this number in the table below:

3	4	.	5	7	1	8
Tens	Ones	Decimal Point	Tenths	Hundredths	Thousandths	Ten Thousandths

Let's practice rounding this number in various ways:

- **Nearest ten or "tens place"**: The digit in the tens place currently is a 3. The digit to the right is a 4. Since this 4 is smaller than 5, keep the 3 as it is, and replace the 4 with a 0. We also get rid of everything after the decimal point. Therefore, the answer in this case is 30.
- **Nearest whole or "ones place"**: This means "round to the ones place." The ones digit is currently a 4. The digit to the right of the ones place is currently a 5. This means that we need to "round up," so the 4 will increase by 1 to become a 5. The rest of the digits after the decimal point go away. Therefore, the answer in this case is 35.
- **Tenths place or "one decimal place"**: The digit currently in the tenths place is a 5. The digit to the right is a 7, which is greater than 5. Therefore, we increase the value in the tenths place from 5 to 6. We then delete everything following the digit in the tenths place. Therefore, the answer in this case is 34.6.
- **Hundredths place or "two decimal places"**: The digit currently in the hundredths place is a 7. The digit to the right is a 1. Since 1 is less than 5, we keep the 7 as is, and delete everything after. Therefore, the answer is 34.57.
- **Thousandths place or "three decimal places"**: The answer is 34.572 Do you see why?

Throughout this book, there will be specific rounding directions on many of the exercises. Two rules also apply throughout, and will be reinforced through examples:

- When computing a number of living things, round to the nearest whole.
- When computing an amount of money, always round to the hundredths place (two digits after the decimal point)

Other rounding directions will be provided as needed.

1.1.4: Exercises

1. Write the number 35,023 using its base 10 expansion, explicitly writing the power of ten correspond to each place.
2. Round the number 67.192 to the following places.
 - a. Tens
 - b. Ones
 - c. Tenths
 - d. Hundredths
3. Consider the following subtraction problem:

$$\begin{array}{r} 374 \\ - 258 \\ \hline \end{array} \quad (1.1.21)$$

Pretend you're explaining to your cousin how to do this by hand, showing her work. What would you say to her? Can you explain what happens when you "borrow?" You should write at least 5-6 sentences and include each step in the process.

4. Go to the link at the start of this section that describes the world population. You will see a website with two tallies, that are counting the World Population and the US population.
 - a. Write down the number that the US population counter says when you look at the website for the first time. It may be slowly increasing, write down the closest number that you can.
 - b. Round that number to the millions place.
 - c. Write a sentence that describes the number you found in the previous part. Your sentence should answer the question "about how many people live in the United States right now?"

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1.2: Scientific Notation

Sometimes we need express very large or very small numbers. For instance, how many cells are there in the human body? And how large, in inches, is a cell? The answers to questions like this can be difficult to understand when expressed in regular notation:

$$\text{Number of cells in the human body} = 37,200,000,000,000$$

$$\text{Length of a cell} = 0.0011811 \text{ inches}$$

For numbers like this, it can be helpful to express them in *scientific notation*.

In this section, you will learn to:

- Recognize and explain scientific notation
- Use scientific notation to compare numbers

1.2.1: Basics of Scientific Notation

The goal of scientific notation is to make numbers shorter to write and easier to compare. It uses two tools you saw in the previous chapter: the base 10 system, and exponents. Let's start with the definition and then see how it is used in a few examples.

Definition: Scientific Notation

A number is written in **scientific notation** when it is expressed as a number between 1 and 10 that is multiplied by a power of 10. As an example, we could write

$$37,200,000,000,000 = 3.72 \times 10^{13} \quad (1.2.1)$$

The exponent on the 10 tells you how many places the decimal point has been moved. In this case, the exponent was moved 13 places to the left.

Scientific notation leverages the base 10 system to write very large or very small numbers in a compact way. In the definition above, we see that a positive exponent moves the decimal point to the left. Think for a moment about why this makes sense, based on the last section. Then try the following example.

Example 1.2.1

Explain why the numbers 37,200,000,000,000 and 3.72×10^{13} are equal.

Solution

Remember from the previous section that exponents represent repeated multiplication, and we can tell a power of 10 by the number of zeroes has following the starting 1. Therefore,

$$10^{13} = 10,000,000,000,000 \quad (1.2.2)$$

When we multiply this number by 3.72, we can see (using either long-hand multiplication, or by thinking about the base 10 expansion), that

$$3.72 \times 10,000,000,000,000 = 37,200,000,000,000 \quad (1.2.3)$$

Therefore, $37,200,000,000,000 = 3.72 \times 10^{13}$

The ideas above will allow us to write any number in scientific notation. The only difference for very small numbers -- those expressed as long decimals -- is that we use negative exponents instead of positive exponents. The following table summarizes how to write both very large and very small numbers in scientific notation.

Number Size	Exponent	Decimal Point Moves...	Example Number	Scientific Notation
Very Large	Positive	Left	453, 000, 000	4.53×10^8
Very Small	Negative	Right	.000000674	6.74×10^{-7}

Let's practice with one more example.

✓ Example 1.2.2

Convert the number .0000049 to scientific notation.

Solution

We see that this is a very small number, between 0 and 1. This means that we need to use a negative exponent to move the decimal point to the right. We need to move the decimal point from its current location to the place between the 4 and the 9, so that the resulting number will be between 1 and 10. If we count places from the current location to the new location, we see that it has to move 6 places. Therefore,

$$.0000049 = 4. \times 10^{-6} \quad (1.2.4)$$

The process of converting from scientific notation to ordinary notation consists of simply reversing the direction that the decimal point travels, and including an appropriate number of zeroes. Let's see two examples of this. Before looking at the solution, see if you can work backwards to figure it out yourself!

✓ Example 1.2.3

Convert each of the following to ordinary notation.

1. 9.362×10^{10}
2. 5.7×10^{-4}

Solution

1. There is a positive exponent here, which means we expect a very large number to result. We will move the decimal point 10 places to the *right*, since we are reversing the process of conversion. We will start putting zeroes once we run out of numbers in 9.362. Therefore,

$$9.362 \times 10^{10} = 93,620,000,000 \quad (1.2.5)$$

2. Here, we have a negative exponent, which means that our result will be a very small number. Since we are reversing the process, we'll move the decimal point 4 places to the *left*, adding zeroes as we go. Therefore

$$5.7 \times 10^{-4} = .00057 \quad (1.2.6)$$

What do "very large" and "very small" mean? As is often the case with topics in this book, the answer depends on the context in which you are using the number. The choice to write 453, 000, 000 versus 4.53×10^8 is up to whomever is writing that number. In general, they should consider the purpose of the number they are writing and who is likely to read it. In the context of a news article, one might see 453,000,000 or perhaps "453 million." The second version of the number gives a sort of alternative scientific notation, in which we've written $453,000,000 = 453 \times 10^6 = 453 \times 1,000,000$. The previous equation does not give the standard scientific notation — which requires that the number in front be between 1 and 10 — but it can be a useful way to express numbers verbally!

However, in a scholarly context, scientific notation is often used in papers and studies to express a variety of numbers, especially when there are multiple numbers of a similar type being compared. The following section shows how scientific notation can be used to compare numbers.

1.2.2: Comparing numbers using scientific notation

One of the benefits of scientific notation is that it allows us to compare the relative size of quantities that are very large or very small. This is especially clear when the quantities are different *orders of magnitude*; that is, when they have different exponents in scientific notation. Using scientific notation can help make this more obvious than ordinary notation.

✓ Example 1.2.4

Which is larger? 89000000 or 9800000

Solution

The most likely way to approach a problem written in this way is to squint at the numbers really hard, and count how many zeroes there are either with your eyes or a tool like a pencil. If you do that, you'll see that there are 6 zeroes in the first number and only 5 in the second number, so the first number must be larger. However, if we had been given these numbers in scientific notation:

$$8.9 \times 10^7 \text{ or } 9.8 \times 10^6 \quad (1.2.7)$$

then it is very clear that the first number is larger, since it is multiplied by a larger power of 10. As a side note, these would be considered quantities with different orders of magnitude; 89000000 is one order of magnitude larger than 9800000.

How might this be useful in context? Consider the following example.

✓ Example 1.2.5

Many chemicals are harmful if ingested, inhaled, or otherwise consumed by humans. Often, we use blood concentrations of chemicals to define the quantity of a substance that is considered harmful. Mercury and lead are two chemicals known to be harmful at very low levels. Lead [can be shown to cause harm](#) at concentrations greater than 3.5×10^{-6} grams per deciliter of blood, whereas mercury [can be shown to cause harm](#) at concentrations greater than 1×10^{-7} grams per deciliter of blood.

Use the information above to compare the effects of equal amounts of lead and mercury exposure.

Solution

There are many ways to answer this question, of course. The point of this example, and of similar exercises you will find throughout the text, is to practice reading, analyzing, and describing quantitative information in context.

What we see in the paragraph above is that while lead and mercury are both dangerous to humans, even at small quantities, a small amount of mercury is more harmful than the same amount of lead, since the concentration required for harm is lower than that of lead.

For example, someone with 1.5×10^{-7} grams of mercury per deciliter of blood would be above the level of toxicity for mercury, but someone with the same amount of lead — 1.5×10^{-7} grams of lead per deciliter of blood — would be below the level of toxicity for lead, since 1.5×10^{-7} is less than 3.5×10^{-6} .

The use of scientific notation allows us to compare these two numbers more clearly than if they were written in standard notation.

And of course, please note: both mercury and lead, at any level, can be very dangerous, and represent serious environmental hazards for many. Please avoid these chemicals as much as possible.

1.2.3: A note about notation

The expression of scientific notation above is common in published textbooks and journals in a variety of fields. However, technology such as spreadsheets, calculators, and computation programs use various shorthand for scientific notation. There are many variations on this as various manufacturers have adopted different conventions, but a fairly common notation is the use of "E" to indicate the power of 10 in an expression in scientific notation. For example, your computer might display the number 587,000,000 as "5.87E8," which is shorthand for 5.87×10^8 . Be aware of this when you are using technology to handle very large or very small numbers!

1.2.4: Exercises

1. Convert 792,000,000,000 to scientific notation.
2. Convert .0000508 to scientific notation.
3. Convert 8.6×10^7 to ordinary notation.
4. Convert 7.2×10^{-9} to ordinary notation.
5. Which is larger: 3.02×10^5 or 4.02×10^4 ? Explain how you know.
6. Which is larger: 6.11×10^{-15} or 1.001×10^{-13} ? Explain how you know.
7. Find a recent news article (from the last 6 months) that contains a very large or very small number. (For the purposes of this exercise, "very large" means greater than 1 million, and "very small" means less than 1 one-millionth.) For that number:
 - a. Write the number in ordinary notation.
 - b. Write the number in scientific notation.
 - c. Of the types of notation — including how the number was originally presented in the article — write a 1-2 sentence reflection on which type of notation you believe most clearly expresses the size of the number and why.
 - d. Include a link to your news article.

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1.3: Ratios

You've likely heard the word "ratio" before — it is perhaps one of the most well-known and commonly use mathematical concepts. In this section we'll define what the word *ratio* actually means, and get practice using ratios to answer questions in various contexts.

 In this section, you will learn to:

- Define, recognize, and compute ratios in various forms
- Compare different ratios
- Use ratios to describe and answer contextualized questions
- Compute two specific real-world ratios

1.3.1: Introduction to Ratios

Humans find it useful to compare quantities. Given two quantities, we might want to compare them by using words like *larger*, *smaller*, *much less than*, *much more than*, and so on. These types of comparisons provide some qualitative information about how two quantities can be related.

Ratios give us a more precise way to compare two quantities. For example, if you're trying to make cookies, your recipe might call for 2 cups of flour and 1 cup of sugar. In this case, you'll use "one more" cup of flour than sugar. But let's say you wanted to make a big batch -- one that used 3 cups of sugar instead of 1 cup. In this case, would you still use one more cup of flour -- in this case, 4 cups of flour?

If you did this — used 4 cups of flour for 3 cups of sugar — your cookies would be *very* sweet and be likely to burn in the oven. The reason is that quantities used in baking are proportional. When you triple the amount of sugar, you also need to triple the amount of flour! That is, you'd need $2 \times 3 = 6$ cups of flour if you plan to use $1 \times 3 = 3$ cups of sugar.

The commonality here is not the difference between the two quantities. Rather, it's the fact that in either case, there is *twice* as much flour as sugar. This is a situation in which a ratio is useful to compare two quantities.

Definition: Ratio

A **ratio** is a way to describe the relative size of two quantities.

There are several common ways in which ratios are represented. We will study three of them.

Name	Notation	Example
Verbal Notation	a to b	A 2 to 3 ratio
Colon Notation	$a:b$	A 2: 3 ratio
Fraction Notation	$\frac{a}{b}$	A $\frac{2}{3}$ ratio

It is important to note that these three things are different ways of expressing the same idea. However, some of them, in particular Fraction Notation, can also have slightly different meanings in other contexts.

Each of these notations has their virtues and their drawbacks. The virtue of using Fraction Notation is that it is standard across all of mathematics. The benefit of this is that there are clear rules for how to manipulate fractions, which you can rely upon to help you solve real world problems. One drawback of Fraction Notation is that using fractions tends to make it look like you are considering part of a whole, which may not always be the case when you are using ratios.

The virtue of Verbal or Colon Notation is that they both make it clear that you are comparing two quantities. For example, we might say: "at Western Oregon University the ratio of out-of-state to in-state students is about 1: 4." The colon makes it clear that for every out-of-state student there are four in-state students. However, if we wrote instead that the ratio is $\frac{1}{4}$, it is tempting to jump

to the conclusion that we are making some claim about one fourth of the student population. But we are not really talking about one fourth of anything! This is why you will see all of these notations represented both in this textbook and in the broader world.

It is possible for two ratios that look different to actually represent the same ratio! That happens if the numbers in one ratio can be multiplied by the same number to get the other ratio. For example, $\frac{2}{3}$ and $\frac{8}{12}$ represent the same ratio, because $2 \times 4 = 8$ and $3 \times 4 = 12$. In other words, we can take the ratio $\frac{2}{3}$, multiply both the top and bottom by 4, and get $\frac{8}{12}$. Thus, they are *equivalent ratios*.

You may also notice that if you *reduce* the fraction $\frac{8}{12}$ by cancelling the common factor of 4 from the top and bottom, you get $\frac{2}{3}$. That's not a coincidence! In general, to compare or easily visualize ratios, it is often helpful to write them in a common way:

Definition: Lowest Terms and Equivalent Ratios

A ratio is in **lowest terms** if its fractional form has been fully reduced. That is, all common whole number factors on the top and bottom of the fraction have been cancelled out.

Two ratios are **equivalent ratios** if, when reduced to lowest terms, they are identical.

Let's see an example of how we can write ratios in different ways, and how to put a ratio in lowest terms.

✓ Example 1.3.1

Write the ratio $\frac{6}{3}$ using two other types of notation. Then write $\frac{6}{3}$ in lowest terms. Finally, write three equivalent ratios in fractional form.

Solution

First, we will write $\frac{6}{3}$ in the two other types of notations. They are:

- 6 to 3
- 6:3

Next, we will reduce $\frac{6}{3}$ to lowest terms. There is a common factor of 3 in both the top and bottom of the fraction, so when we cancel those factors out, $\frac{6}{3}$ reduces to $\frac{2}{1}$. Now there are no more common factors (besides 1, but canceling that does not change the answer). So, this ratio in lowest terms is $\frac{2}{1}$. Note that this is probably more easily understood than $\frac{6}{3}$ in most contexts.

Last, we'll find several equivalent ratios. To do that, we'll multiply both the top and bottom of the fraction $\frac{6}{3}$ by the same numbers. For example, if we multiply both by 2, we get $\frac{12}{6}$. This is equivalent to $\frac{6}{3}$. We also found that $\frac{2}{1}$ is equivalent to $\frac{6}{3}$, since we can multiply both the top and bottom by $\frac{1}{3}$. Said another way, you can see this by dividing both the top and bottom numbers by 3.

There are many other equivalent ratios: $\frac{4}{2}$, $\frac{10}{5}$, $\frac{18}{9}$, $\frac{200}{100}$ — any fraction where the top number is twice the bottom number. This is the essence of equivalent ratios: even though they have different numbers, those two numbers have the same relationship to one another.

Next, we need to understand ratios in context. See if you can answer the question below before revealing the answer!

✓ Example 1.3.2

The director of an animal shelter is writing a newsletter about the current operations at the shelter. In the animal shelter, there are 18 dogs and 24 cats. In her newsletter, she wants to list the ratio of cats to dogs in an easy-to-understand way. What sentence should she write?

Solution

Here's one possibility: the ratio of cats to dogs in the shelter is 24:18. (Notice that the order matters here!)

That ratio gives an accurate description, but it doesn't lead to an easily visualized comparison for the newsletter readers. So, she should put it in lowest terms. If we rewrite our ratio as $\frac{24}{18}$, we can reduce this to lowest terms by eliminating a factor of 6 from both top and bottom. Then the ratio becomes $\frac{4}{3}$, which has no more common factors to cancel.

The sentence she writes will be: "There's a 4 to 3 ratio of cats to dogs in the shelter."

Now, of course, there are many possible good answers to the question above. In this class, grading will be much more fluid than in other math classes, because often there is more than one good answer when we are asked to contextualize mathematics. You should focus on making the mathematics understandable in context. Throughout these notes you will see many more examples of how to do that.

In some contexts, ratios are rounded slightly so they are easier to understand. This happens particularly when the actual quantities being compared are large, but it's helpful to represent the ratio using smaller numbers.

✓ Example 1.3.3

According to the [WOU website](#), as of 2023, the total enrollment is listed as 3752 students, and the student-to-faculty ratio is listed as 13: 1. You want to know how many faculty there are at WOU. How might you find this out, and how do you explain your answer?

Solution

To figure this out, we need to find a student-to-faculty ratio that is equivalent to 13: 1, but where the number of students is 3752. We will use the a letter, x , to represent the number of faculty. That is, we want to find a value for x so that

$$3752 : x \quad \text{and} \quad 13 : 1 \quad (1.3.1)$$

are equivalent ratios. If these are equivalent ratios, then we must be able to multiply 13 by some number to get 3752. To find this number, we simply divide $3752 \div 13$ since multiplication and division are inverse operations. This gives us

$$3752 \div 13 = \frac{3752}{13} \approx 288.6 \quad (1.3.2)$$

using a calculator, rounding to the nearest tenth. That is, $13 \times 288.6 \approx 3752$ You can check this using a calculator.

In order for these ratios to be equivalent, we have $1 \times 288.6 = x$ Therefore, it must be $288.6 = x$; and so there are 288.6 faculty at WOU. But this answer is not a whole number, even though it is referring to a number of people. So, what does it mean to have 288.6 faculty?

It could mean several things. First, it could mean that some faculty are part-time, which means that they are counted as only part of a whole faculty member. But it could also mean that the ratio 13: 1 was just slightly rounded, which is also likely. For example, say there are 289 faculty — then the real student to faculty ratio would be 3752: 289 which cannot be reduced. This ratio, 3752 to 289, would not be a good ratio to put on the website, because it's hard to understand. The ratio 13: 1, while it may have been rounded just slightly, provides a better picture of what class sizes at Western look like.

Now, in the previous example, you may have seen a quicker way to solve this problem using proportions. In general, if you find a way to solve a problem that works for you, as long as you get the same numerical answer, you can use that method. We will cover proportions in the next section. In this section, we are choosing to focus on equivalent ratios as a method for problem-solving.

Let's see one more example.

✓ Example 1.3.4

In a given afternoon, a librarian checks out 52 total books. Of those, 24 are fiction books. What is the ratio, in lowest terms, of non-fiction books to fiction books checked out of the library that afternoon?

Solution

In a given afternoon, a librarian checks out 52 total books. Of those, 24 are fiction books. What is the ratio, in lowest terms, of non-fiction books to fiction books checked out of the library that afternoon?

In this problem we need to know that library books are either fiction or non-fiction books. Therefore, the number of non-fiction books can be found by calculating $\frac{52}{\text{total books}} - \frac{24}{\text{fiction books}} = \frac{28}{\text{non-fiction books}}$

Thus, the ratio of non-fiction books to fiction books is 28:24. We see that both 28 and 24 have a factor of 4, so when we divide both parts of the ratio by 4, we get a ratio of 7:6. That ratio is in lowest terms, so the answer is: There was a ratio of 7:6 non-fiction books to fiction books checked out of the library that afternoon.

1.3.2: Two Important Real-World Ratios

The remainder of this section is dedicated to two real-world ratios that you will likely encounter at some point in your life. They are:

- Your *cholesterol ratio*
- Your *debt-to-income ratio*

Before we proceed, you should know: the person writing these notes is neither a medical doctor nor a financial professional. The information in this section should not replace the advice of an actual physician or financial planner! The guidelines outlined below are based on publicly available information, and sources are cited where appropriate. In some cases some simplifications have been made for the sake of clarity.

First, we'll study the cholesterol ratio, which is used to indicate risk of heart attack and coronary disease by doctors. Cholesterol is a fat-like substance that's found in many common foods. There are two main types of cholesterol:

- **HDL**, or **high-density lipoprotein**, which is sometimes called "good cholesterol." HDL carries other cholesterol away from your arteries and to your liver, thereby reducing the risk of clogged arteries. HDL is found in foods like avocado, nuts, and vegetable-based oils. Smoking can also decrease HDL levels.
- **LDL**, or **low-density lipoprotein**, which is sometimes called "bad cholesterol." LDL is the type of cholesterol that sticks to the sides of your arteries, thereby increasing the risk of clogged arteries. LDL is found in foods with lots of saturated and trans fats, like butter, red meat, and fried foods. Smoking can also increase LDL levels.

You can get your own cholesterol tested by requesting a lipoprotein panel from a physician. In general, [it is ideal](#) to have an LDL level below 100 milligrams per deciliter (mg/dL) and an HDL level above 45 mg/dL. Additionally, your total cholesterol should be below 200 mg/dL. Various demographic factors affect your personal ideal levels.

However, since the two types of cholesterol interact, doctors will often monitor a number known as your *cholesterol ratio*. This is calculated in the following way:

$$\frac{45}{\text{HDL}} + \frac{110}{\text{LDL}} = \frac{155}{\text{total cholesterol}} \quad (1.3.3)$$

where the "total cholesterol" is calculated as

$$\text{cholesterol ratio} = \frac{\text{total cholesterol}}{\text{HDL}} = \frac{155}{45} \approx 3.44 \quad (1.3.4)$$

Since HDL is good cholesterol, and LDL is bad cholesterol, a low cholesterol ratio is an indicator of better health and lower heart attack risk. Mathematically, we want the top of the fraction, LDL + HDL, to be relatively low compared to the bottom, which is just HDL. **In general, the ideal ratio is 3.5 or lower, and should be kept below 5.**

Note that the ratio has no units — it is simply a number. As with LDL and HDL recommendations, various demographic factors affect the exact recommended levels. You can find more information about this [here](#).

✓ Example 1.3.5

You receive the results of a lipoprotein panel from your doctor. Your LDL is 110 mg/dL, and your HDL is 45 mg/dL. What is your cholesterol ratio, and what does this indicate about your risk for heart disease? Round numerical answers to two decimal places.

Solution

To calculate your cholesterol ratio, we need to know the total cholesterol. We calculate this as follows:

$$\frac{45}{\text{HDL}} + \frac{110}{\text{LDL}} = \frac{155}{\text{total cholesterol}} \quad (1.3.5)$$

Now we calculate your cholesterol ratio as follows

$$\text{cholesterol ratio} = \frac{\text{total cholesterol}}{\text{HDL}} = \frac{155}{45} \approx 3.44 \quad (1.3.6)$$

This ratio represents a fairly low risk for heart disease, since it is below 3.5. Additionally, your total cholesterol is below 200 mg/dL, which is good. However, the LDL number is a bit high — ideally, it should be below 100 mg/dL.

The last ratio we'll study is the *debt-to-income* ratio, also called *DTI*. This ratio is used frequently by lenders to determine if someone is eligible to borrow money for a large purchase or credit line. This is not be confused with a person's credit score.

To calculate a person's debt-to-income ratio, we use the following formula:

$$\text{DTI} = \text{debt-to-income ratio} = \frac{\text{total monthly debt payments}}{\text{monthly gross income}} \quad (1.3.7)$$

Let's examine the two parts of the fraction. A person's *monthly gross income* is the amount of money they make in a month before any taxes or deductions, such as federal tax, state tax, or Social Security tax are taken out. This amount can be found on a pay stub, and it's likely higher than what you actually take home in your paycheck.

A person's *total monthly debt payments* can be calculated by **adding up the debt payments** they make each month to debts they have. Debt payments include things like:

- House payment (mortgage) or rent payments
- Student loan payments
- Auto loan payments
- Credit card payments
- Child support or alimony payments
- Any other type of debt: personal loan, medical loan, etc.

Other monthly payments that you might make are typically not considered debt. For example, groceries, utilities, subscriptions, and gas do not count towards your total monthly debt payments

It's important to note that different lenders use slightly different definitions of what counts as debt. The list above is intended to be as general as possible, and is what we will use in this class; however, you may encounter slightly different criteria when you actually apply for a loan. For more information, visit a financial institution's website such as [this one](#).

Once the debt-to-income ratio is calculated, it is compared to a benchmark number. **In practice, that benchmark number varies, but is typically around .36, which is what we will use in this class.** If your DTI is less than .36, that means you are a good candidate for lending — your debts are relatively low compared to your income. If your DTI is more than .36, that means you are a riskier candidate for lending, because your debts are already somewhat high compared to your income.

In general, you want to keep your debt-to-income ratio as low as possible. However, some lenders accept a DTI of up to 50%, depending on the type of loan.

✓ Example 1.3.6

You currently make \$2800 gross income per month. Your rent costs \$550 per month, your monthly minimum credit card payments are \$300, you spend \$350 per month on groceries, and your car payment is \$150 per month. What is your debt-to-income ratio? If you applied for a home loan, do you think you would be approved? Why or why not?

Solution

First, we add up all of the monthly debt payments. Be careful -- not every monthly expense is a debt payment! You will need to read carefully and consult the list of debts above to tell what expenses are and aren't debts. The debts in this case are the rent, credit card payments, and car payments. The groceries are not a debt payment, so we don't include them. We add the debt payments to find the total monthly debt payments:

$$\underset{\text{rent}}{\$550} + \underset{\text{credit cards}}{\$300} + \underset{\text{car loan}}{\$150} = 1000 = \text{total monthly debt payments} \quad (1.3.8)$$

Now, to find the DTI, we calculate

$$DTI = \frac{\text{total monthly debt payments}}{\text{Monthly gross income}} = \frac{\$1000}{\$2800} \approx .357 \quad (1.3.9)$$

We see that this DTI is just slightly less than the ideal number, .36. That means that you're likely a good candidate for a home loan, but close to being a risky candidate. You are fairly likely to be approved for a home loan.

1.3.3: Exercises

Make sure to include supporting work for all answers

1. Consider the ratio 6: 11.
 - a. Write 6: 11 in two other ways.
 - b. Write three equivalent ratios.
 - c. Is the ratio 6: 11 in lowest terms? Explain why or why not in a complete sentence.
2. In a given pond, there are 22 rock fish for every 121 minnows. Express the ratio of minnows to rock fish in lowest terms.
3. The ratio of the amount of food that my cat Jade eats to the amount of food that my cat Poe eats is 3: 2. If Jade eats 6 ounces of food, how many ounces of food will Poe eat? (Your answer should be a whole number.)
4. Sourdough bread uses a 5 to 4 ratio of flour to sourdough starter. If you have 10 cups of flour, how much sourdough starter should you use?
5. Elizabeth's total cholesterol is 182 and her LDL is 128. What is her cholesterol ratio? How might you evaluate her overall heart health in 1-2 sentences?
6. Mark's gross monthly income is \$2300. His rent is \$575, his monthly car payment is \$120, his utilities cost \$300 per month, his student loan payment is \$63 per month, and he spends \$300 per month on groceries. What is his debt-to-income ratio? If he applied for a mortgage, do you think he would be approved? Why or why not?
7. Do a bit of research and write a paragraph (4-5 sentences) about what a person's *credit score* is, including how it is calculated and how it affects one's chances of being approved to borrow money.

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1.4: Proportions

In the previous section, we learned that a *ratio* is a comparison of two quantities. However, many of the problems we solved involved comparing multiple ratios, and often required finding an equality between two ratios. These sorts of problems are most easily solved using *proportions*, which are the subject of this section.

 In this section, you will learn to:

- Recognize and set up proportion problems
- Apply the Cross Multiplication and Division undoes Multiplication methods to solve proportion problems
- Use the Compare to the Whole method to solve problems involving proportions

1.4.1: Proportions: Definition and Basic Methods

Let's recall our starting example from the previous section, in which we had a recipe that called for 2 cups of flour and 1 cup of sugar. We asked the question: if we wanted to make a larger version of the same recipe using 3 cups of sugar, how much flour should we use?

We know now that ratios can be expressed as fractions, and whenever ratios are equivalent, the fractional representations of those ratios are equal. So, let's call the number of cups of flour x and set up the following equality:

$$\frac{2 \text{ cups of flour}}{1 \text{ cup of sugar}} = \frac{x \text{ cups of flour}}{3 \text{ cups of sugar}} \quad (1.4.1)$$

We need to figure out the value of x that makes this equation true. That is, what number can be substituted in place of x so that these fractions truly are equal? In this case, the easiest way is to guess-and-check, which is a perfectly valid solution method with small numbers. If you just try some small numbers, you can find that $x = 6$ is the solution, because $\frac{6}{3}$ reduces to $\frac{2}{1}$, and thus $\frac{6}{3}$ and $\frac{2}{1}$ are equivalent ratios.

In this chapter, we'll learn how to solve problems like this generally, including those that can't be solved using guess-and-check. Our primary tool will be a *proportion*.

Definition: Proportion

A **proportion** is an equality between ratios.

This means that, in a strictly mathematical sense, a proportion is an equation. For example,

$$\frac{5}{2} = \frac{40}{16} \quad (1.4.2)$$

is a proportion, because it contains two ratios that are equal to one another. You may have heard the word proportion used in other ways, and that's because the word "proportion" has a different mathematical meaning than its typical English usage. That doesn't mean either usage is wrong; rather, it is dependent upon context. In this book, we will use the word *proportion* to mean any equation that looks like this:

$$\frac{a}{b} = \frac{c}{d} \quad (1.4.3)$$

where a , b , c , and d will usually be numbers or variables.

The reasons we care about proportions is that they give us a way to *find an unknown part of one of the ratios involved*. Recall the following example, which we saw in the Ratios section:

✓ Example 1.4.1

On the [Western Oregon University website](#), the total enrollment is listed as 3752 students, and the student-faculty ratio is listed as 13: 1. You want to know how many faculty there are at WOU. How might you find this out, and how do you explain your

answer?

In solving this problem before, we set up two ratios

$$3752:x \quad \text{and} \quad 13:1 \quad (1.4.4)$$

Why did we do this? Well, it turns out that all proportion problems can be solved using a method from algebra known as *cross multiplication*. While this text mostly stays away from algebra, this procedure is essential. The good news is that it works the same way every time, and it's not very complicated.

Cross Multiplication

If you have a proportion of the form:

$$\frac{a}{b} = \frac{c}{d} \quad (1.4.5)$$

then "cross multiplication" refers to rewriting the equation in the following equivalent way:

$$c \times b = a \times d \quad (1.4.6)$$

In other words, we are "crossing" from a to d in the $\searrow$ direction and from c to b in the $\swarrow$ direction.

A quick mathematical note: what we're really doing is multiplying both sides by b and d , and then canceling common factors -- but calling it cross multiplication seems to make it easier for students to understand and remember.

Let's practice cross multiplying in our example. We had the proportion:

$$\frac{3752}{x} = \frac{13}{1} \quad (1.4.7)$$

Cross multiplying this gives us the following equation:

$$13 \times x = 3752 \times 1 \quad (1.4.8)$$

What good did that do? Well, note that we can simplify a little bit. It's typical to omit the $\times$ when a variable is multiplied by a number, so we can rewrite $13x$ for $13 \times x$. We also know that $3752 \times 1 = 3752$ So our equation becomes

$$13x = 3752 \quad (1.4.9)$$

Now what? We've eliminated the fractions, but we can't yet say what x is. In order to find x , we need one more algebra procedure, which we will call *Division undoes Multiplication*.

Division undoes Multiplication

Given an equation of the form

$$Ax = B \quad (1.4.10)$$

where A and B are numbers, we can find the value of x by dividing both sides by A . That is,

$$x = \frac{B}{A} = B \div A \quad (1.4.11)$$

Once again, we are using properties of fractions here: mathematically, we are dividing both sides by A and then reducing:

$$\begin{aligned} Ax &= B \\ \frac{Ax}{A} &= \frac{B}{A} \\ \cancel{A}x &= \frac{B}{\cancel{A}} \\ x &= \frac{B}{A} \end{aligned}$$

But this is another procedure used so frequently that it's worth giving it a name.

Back to our example: we had the equation

$$13x = 3752 \quad (1.4.12)$$

We now have a tool to find x -- the fact that *Division undoes Multiplication*! Using this procedure, we have

$$x = \frac{3752}{13} = 3752 \div 13 \approx 288.6 \quad (1.4.13)$$

This is the same answer we found before, but we used a slightly different method. And keep in mind that, just as the previous section, we would need to round this answer to 289 faculty to make sense in context. That said; the main point is now we now have a fool-proof way to solve this type of equation!

While this may seem more complicated at first, you'll find that the following sequence of steps will always work to solve proportions:

Solving Proportions

- Set up the proportion with exactly one unknown value, called x .
- Apply the Cross Multiplication.
- Apply Division undoes Multiplication.

We will get lots of practice with this procedure in the exercises for this section. Once you practice with the procedures above, you'll find that it's not too bad. The hardest part is often the first step — setting up the proportion correctly. That's the part that depends on reading the question very carefully! In general, the way to set up a proportion involves keeping track of units. Let's see an example to understand.

✓ Example 1.4.2

In an office supply store, 8 markers cost a total of \$12.00. Assuming all markers are equally priced, how much would 6 markers cost?

Solution

This is a problem that is suitable to be solved using proportions because the markers are all equally priced, meaning that the ratio of *total cost* : *number of markers purchased* will be the same, no matter how many markers are purchased. That means we can set up the following proportion:

$$\frac{12 \text{ dollars}}{8 \text{ markers}} = \frac{x \text{ dollars}}{6 \text{ markers}} \quad (1.4.14)$$

Notice how, in the equation above, we are *labeling the units* of all quantities involved. Moreover, the units on each side match: dollars are on top, markers are on bottom, and the corresponding quantities are grouped on each side of the equation -- \$12 for 8 markers, and \$ x for 6 markers. Labeling your units in this way will help you avoid mistakes with units!

Now that we've gotten our proportion set up correctly, we can rewrite it without labels:

$$\frac{12}{8} = \frac{x}{6} \quad (1.4.15)$$

From here, we'll follow the last two steps: cross multiply, and then use division to find x . Using Cross Multiplication, we have

$$x \times 8 = 12 \times 6 \quad (1.4.16)$$

On the left, we can rewrite $x \times 8$ as $8x$, since multiplication can always switch orders. Then we can simplify to get

$$8x = 72 \quad (1.4.17)$$

Now we can use Division undoes Multiplication to get

$$x = \frac{72}{8} = 72 \div 8 = 9 \quad (1.4.18)$$

Therefore, $x = 9$. Now, we want to make sure our answer actually means something. What are the units on x ? Well, if we look back at our original proportion,

$$\frac{12 \text{ dollars}}{8 \text{ markers}} = \frac{x \text{ dollars}}{6 \text{ markers}} \quad (1.4.19)$$

we see that x is a number of dollars. Thus, we can say that $x = \$9$, which means that 6 markers will cost \$9.

You may be thinking: there is a much faster way to do that! And that may be true for you. Once again, the point is not to mimic a particular method for problem solving here — these notes will show some good ways of solving a problem, but they cannot cover every good solution. They are intended to highlight themes and strategies that will work for many types of situations. Other ways you may have solved the problem above include:

- Calculate the cost per marker to be \$1.50, and multiply that number by 6 markers to get \$9.
- Calculate that 6 is $\frac{3}{4}$ of 8, so the cost of 6 markers would be $\frac{3}{4}$ the cost of 8 markers, and $\frac{3}{4}$ of \$12 is \$9.
- Set up a different initial proportion, such as $\frac{12 \text{ dollars}}{x \text{ dollars}} = \frac{8 \text{ markers}}{6 \text{ markers}}$ or $\frac{8 \text{ markers}}{12 \text{ dollars}} = \frac{6 \text{ markers}}{x \text{ dollars}}$ and then solved that proportion.

What's amazing about the last point above is that both of those proportions — which were different than the method used in the solution above — still give the same answer! This shows that there are *many* different ways of approaching the same problem. All you need to do is find the one that works for you, and be able to explain your work.

1.4.2: Comparing to the Whole

Sometimes a problem involving proportions will be less straightforward. For example, consider the following:

✓ Example 1.4.3

In a rainforest in Panama, the ratio of two-toed sloths to three-toed sloths is 10: 3. There are 741 total sloths in the rainforest. How many of them are two-toed?

In the problem above, we are given one ratio that compares the quantity of two-toed versus three-toed sloths. However, we are not given any information about the actual numbers of either two- or three-toed sloths. We simply know the comparison between them. Instead, we are just given the total number of sloths, but no actual breakdown into how many fall into each category. How are we supposed to find the number of two-toed sloths from just this information? We can't readily write down a proportion like we were able to in the previous example, because the units would be wrong; we need to compare like quantities. This situation calls for one more procedure.

Compare to the Whole

Assume there are two quantities, x and y , neither of which you know. However, you know two things about them

- The total $x + y$ (the total number of both quantities)
- The ratio of quantity x to quantity y is $a: b$

Then you can use the Compare to the Whole method. This says that, to find quantity x , you use the proportion

$$\frac{a}{a + b} = \frac{x}{x + y} \quad (1.4.20)$$

and then find x . Note: you already know $x + y$, since it is the total number of both quantities.

Let's see how this procedure can be applied to the sloth example.

✓ Example 1.4.3 Revisited

In a rainforest in Panama, the ratio of two-toed sloths to three-toed sloths is 10: 3. There are 741 total sloths in the rainforest. How many of them are two-toed?

Solution

In this question, our two quantities x and y are the number of two- and three-toed sloths, respectively. We are asked to find x , the number of two-toed sloths. Our known ratio is 10: 3, so using the notation of the Compare to the Whole method, we have $a = 10$ and $b = 3$, and $a + b = 13$. We also know that the total number of sloths is 741, so $x + y = 741$. So we'll set up the following proportion -- pay close attention to the labels!

$$\frac{10 \text{ two-toed sloths}}{13 \text{ total sloths}} = \frac{x \text{ two-toed sloths}}{741 \text{ total sloths}} \quad (1.4.21)$$

On the righthand side of the proportion above, the ratio $\frac{x}{741}$ represents the actual number of sloths, in which there are x two-toed sloths out of a total of 741 total sloths.

On the lefthand side, the ratio $\frac{10}{13}$ represents an imaginary "smaller but proportional rainforest," in which there are only 10 two-toed and 3 three-toed sloths, for a total of 13 sloths in our imaginary smaller rainforest.

Proportionality says that these proportions must be equal, but since we don't know the breakdown of the total number of sloths, we must *compare to the whole*, which means we must compare the total number of sloths on each side. We get a total number of 13 on the left by computing $10 + 3$, and on the right, we know the total to be 741.

Once we have that proportion, we can simply solve it using our processes from the previous section. From the proportion

$$\frac{10}{13} = \frac{x}{741} \quad (1.4.22)$$

we use *Cross Multiplication* to obtain

$$13x = 7410 \quad (1.4.23)$$

and then use *Division undoes Multiplication* to get

$$x = \frac{7410}{13} = 570 \quad (1.4.24)$$

Looking back, we see that x represents the number of two-toed sloths. Therefore, there are 570 two-toed sloths in the rainforest.

That's the best way to think about the Compare to the Whole method -- the ratio you are given represents a "smaller version" of the situation described, and to find the total quantity in the smaller version, you simply add the two parts together. Then compare that to the actual total quantity using a proportion. *If the problem statement contains words like "total," "whole," or "all together," it's likely that you'll need to use the Compare to the Whole method.* However, as always, the most important thing is to read the problem and think critically about what it's asking!

P.S. for this section: You may notice that some of the algebra is becoming less explicit as we see more and more examples. If you are confused about why an algebra or arithmetic step is true, try looking for a similar problem earlier in this book — there is likely an explanation there. If you can't find one, or are still confused, you should ask your instructor or email the author of this book at merrill@wou.edu.

1.4.3: Exercises

When you are completing these exercises, make sure to show supporting work.

1. You can walk 2 miles in 36 minutes. How long will it take you to walk 5 miles? Give your answer as a number of hours plus a number of minutes (that is, you would express 70 minutes as "1 hour and 10 minutes"). Remember that there are 60 minutes in an hour!
2. You can mow $\frac{1}{3}$ of an acre of lawn in 90 minutes. How long would it take you to mow 2 acres of lawn? Give your answer as a number of hours.
3. When brewing an amber ale (a type of beer), recipes typically call for an 8:2 ratio of pale malts to crystal malts (these are types of grain in the beer). If you are brewing a 10 gallon batch of amber ale, you need a total of 22 pounds of malt. How many pounds of each type of malt (pale and crystal) should you buy? Make sure to indicate both answers clearly, and do not round them -- decimals are fine. [Hint: they should add up to 22 pounds!]
4. The ratio of the length of the shadow cast by an object to the height of that object is 2-to-3 at a certain time of day.
 - a. How long of a shadow does a 6 foot tall person cast?
 - b. If a shadow of a tree is 20 feet long, how tall is the tree?
5. The ratio of registered Democrats to registered Republicans is 47 : 52 in Polk County. There is a total of 8920 registered Democrat and Republican voters. How many of them are Democrats?
6. The following is an excerpt* of [an article](#) about sea turtle sex ratios (e.g., what proportion of the population is female or male.):

"More than 200,000 sea turtles nest on or near Raine, a tiny 80-acre curl of sand along the northern edge of the [Great Barrier Reef](#), the portion hardest hit by warming waters. The other portion of that sea turtle population nests further from the equator, near Brisbane, where temperature increases have not been as dramatic.

What Allen and Jensen discovered was significant. Older turtles that had emerged from their eggs 30 or 40 years earlier were also mostly female, but only by a 6 to 1 ratio. But younger turtles for at least the last 20 years had been [more than 99 percent female](#). And as evidence that rising temperatures were responsible, female turtles from the cooler sands near Brisbane currently still only outnumber males 2 to 1.

Six weeks after Allen and Jensen published their results, [another study](#) from Florida looking at loggerheads revealed that temperature is just one factor. If sands are moist and cool, they produce more males. If sands are hot and dry, hatchlings are more female.

But new research in the last year also offered rays of hope."

- a. What ratios can you find above? Write them down, stating explicitly what they are comparing.
- b. What two factors does this article assert affect the sex of sea turtles? List them.
- c. Given the information in the article, if a randomly selected group of 120 turtles from Brisbane have their sex examined, how many do you expect to be female? Show your work.
- d. Write a 2-4 sentence reaction to the article excerpt above, and make sure to answer the following question: do you feel that that ratios in the article are presented in a way that makes sense? If not, how else could you present this same information?

[*Note: you can access the article for free if you enter your email when prompted; however, you do NOT need to access the article answer this question.]

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CHAPTER OVERVIEW

2: Numbers in Context

Explore how relationships between numbers shape our everyday life and learn skills to answer common questions involving numbers.

[2.1: Types of Proportionality](#)

[2.2: Rates](#)

[2.3: Percents](#)

[2.4: Applications of Percents](#)

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2.1: Types of Proportionality

In the previous section, we studied proportions, and used them to solve problems involving ratios. In this section, we continue our study of proportions, and investigate two different types of proportionality.

 In this section, you will learn to:

- Recognize direct proportionality relationships, and use them to answer questions involving direct proportionality
- Compute and interpret the constant of proportionality in context
- Recognize inverse proportionality relationships, and use them to answer questions involving inverse proportionality

There are two main types of proportionality. We will learn about the more common one first.

2.1.1: Direct Proportionality

 Definition: Directly Proportional

Two quantities are **directly proportional** if, as one quantity increases, the other quantity also increases at the same rate.

Direct proportionality describes all of the proportion problems we've seen before. Here is another example that shows how direct proportionality works, and introduces the next important notion.

✓ Example 2.1.1

At an hourly wage job, you work for 5 hours, and get paid \$83.75. How much money will you earn if you work 7 hours? (Assume that you are not making overtime pay, or any other sort of special pay rate.)

Solution

This is a situation described by direct proportionality. Since this is an hourly wage job, and we're told that there is no overtime pay or other special pay rate, we can safely assume that we make the same amount per hour in this job. In other words, the rate of pay will be the same no matter how many hours we work. That means that if the number of hours worked increases, the amount you're paid increases at the same rate, no matter the number of hours worked.

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We can solve this using a proportion, using the same techniques as in the last section. We will set up the following proportion equation:

$$\frac{\$83.75}{5 \text{ hours}} = \frac{\$x}{7 \text{ hours}} \quad (2.1.1)$$

Notice that we have picked a variable, x , to denote the answer we are trying to find -- the number of dollars earned for working 7 hours. Also notice how we've labeled our units, and have made sure that the corresponding quantities are together. If you do this every time -- label your units, and make sure corresponding quantities stay together -- you can solve any direct proportion problem.

Now for the process that will actually help us solve for x . First, rewrite the equation without labels:

$$\frac{83.75}{5} = \frac{x}{7} \quad (2.1.2)$$

Next, apply Cross Multiplication

$$5x = 83.75 \times 7 \quad (2.1.3)$$

Next, simplify the right side (using a calculator):

Next, simplify the right side (using a calculator):

$$5x = 586.25 \quad (2.1.4)$$

Finally, apply Division undoes Multiplication to find x :

$$x = \frac{586.25}{5} = 117.25 \quad (2.1.5)$$

That means that if you work 7 hours, you will make \$117.25. Think for a moment to see if that's reasonable: it's more, but not too much more, than you made working for 5 hours. So, it seems like a sensible answer.

The approach above works just fine to find the desired answer. However, what if you wanted to know how much you'd make working for 3 hours? Or 4 hours? Or 10 hours? You could just reproduce the work above each time, you wanted. But you may also find the following approach quicker:

Definition: Constant of Proportionality

In a situation involving directly proportional quantities, the **constant of proportionality** is the common ratio that describes the comparison of any two corresponding quantities. In other words, it is the constant rate of change between the two quantities.

Let's see how to find a constant of proportionality, and how to interpret it.

✓ Example 2.1.1 Revisited

You're in the same situation as the previous example: you work for 5 hours, and earn \$83.75. What is the constant of proportionality in this example, and what does it mean in context?

Solution

The constant of proportionality is the common ratio that describes the comparison of any two corresponding quantities. In this situation, we actually have two sets of corresponding quantities, one of which we found in the previous example. You know that you'll make \$83.75 working for 5 hours, and we calculated above that you'll make \$117.25 if you work for 7 hours. Let's look at these two ratios:

$$\frac{\$83.75}{5 \text{ hours}} \quad \text{and} \quad \frac{\$117.25}{7 \text{ hours}} \quad (2.1.6)$$

Both of these ratios are fractions, and we can simply divide the top by the bottom to reduce them to a single number. Using a calculator, we can see that

$$\frac{\$83.75}{5 \text{ hours}} = \$83.75 \div 5 \text{ hours} = \$16.75 \text{ per hour} \quad (2.1.7)$$

and

$$\frac{\$117.25}{7 \text{ hours}} = \$117.25 \div 7 \text{ hours} = \$16.75 \text{ per hour} \quad (2.1.8)$$

These are the same answer! Do you see why? We originally found our value for x in the previous example by setting the two ratios equal. So, they must give the same answer when divided.

This shared rate — \$16.75 per hour — is the *constant of proportionality* in this situation. It is the shared value of all ratios described by this problem, where the top of the ratio is money earned, and the bottom is hours worked.

What does this constant of proportionality mean in this situation? In this case, the constant of proportionality is your hourly pay rate. *In other words, it's how much you make per hour.*

A few more comments on the example above: now that you know this rate, it's quite simple to find how much money you'll make if you work 3, 4, or 10 hours. You just multiply your hourly rate by the number of hours worked. For example, if you worked 4 hours, you could calculate:

$$\frac{4}{\text{hours}} \times \$16.75 \text{ per hour} = \$67.00 \quad (2.1.9)$$

This means you would make \$67.00 working for 4 hours. Notice that if we reverse the process — in other words, if we try to extract the constant of proportionality knowing that we make \$67.00 in 4 hours, we get:

$$\frac{\$67.00}{4 \text{ hours}} = \$67.00 \div 4 \text{ hours} = \$16.75 \text{ per hour} \quad (2.1.10)$$

It's the same rate we found before. This is why it's called a *constant* of proportionality: it stays the same, even as the corresponding quantities change.

Constants of proportionality will change in meaning depending on the context of the problem. For example, you might ask: If 5 people eat a total of 10 slices of pizza, how many slices does each person eat? In this case, the constant of proportionality could be found by:

$$\frac{10 \text{ slices}}{5 \text{ people}} = 2 \text{ slices per person} \quad (2.1.11)$$

In this case, a correct interpretation would be: "The constant of proportionality is 2 slices per person, which means that each person eats 2 slices of pizza." When asked for an interpretation, you should write a sentence similar to the previous -- your goal is the explain the meaning of the constant of proportionality in context of the situation. You will need to read carefully and use critical thinking to deduce a meaningful interpretation. As with many questions in this class, there are multiple good answers to these types of questions!

2.1.2: Inverse Proportionality

Let's start this section with a question to illustrate the main concept. Before reading ahead, try to answer this question on your own:

✓ Example 2.1.2

Suppose it takes 6 sanitation workers, all working simultaneously, 4 hours to pick up the trash and recycling in a given neighborhood. How many sanitation workers would it take to pick up the trash and recycling in the same neighborhood in 3 hours? (Assume that all sanitation workers work at the same rate, and can work independently.)

Did you try to answer the question yourself? What did you come up with? If you're like many students who carefully read the previous sections, you might have written this down:

$$\frac{6 \text{ sanitation workers}}{4 \text{ hours}} = \frac{x \text{ sanitation workers}}{3 \text{ hours}} \quad (2.1.12)$$

Then you'd apply Cross Multiplication to get:

$$4x = 18 \quad (2.1.13)$$

and then you'd use Division which undoes Multiplication to get

$$x = \frac{18}{4} = 4.5 \quad (2.1.14)$$

Now, the numerical answer 4.5 is a bit nonsensical, because it's talking about a number of people. So you would round up to 5, and say "It would take 5 sanitation workers to pick up the trash and recycling in 3 hours."

But wait a second: this does not make sense! Think about it: if it takes 6 workers 4 hours to accomplish this task, shouldn't it take 5 workers *more* time than 4 hours? After all, there is the same amount of work to be done, but fewer people to do it! So the answer "5 workers" cannot possibly be correct. We expect a number of workers that is *larger* than 6 to get the task done in a shorter amount of time.

We can learn two things from the previous discussion:

- It is important to evaluate whether or not an answer to a question makes sense in context by asking: What sort of answer would I expect to get? Does my answer seem reasonable?

- Not all problems can be solved using direct proportionality!

The good news is that this type of problem can be solved in a relatively simple way. We define the main concept in this section to see how these problems work.

Definition: Inversely Proportional

Two quantities are **inversely proportional** if, as one quantity increases, the other quantity decreases at the same rate.

Note how similar this definition is to the previous definition of direct proportionality. The only difference here is that one quantity increases while the other decreases:

Direct Proportionality	Inverse Proportionality
As one quantity <i>increases</i>	As one quantity <i>increases</i>
the other quantity <i>increases</i>	the other quantity <i>decreases</i>

In order to determine what type of problem you're working on, you'll need to think critically about the quantities involved, and use clues from your experience and the context of the problem to determine how the quantities are related. Things like the previous problem — when a group of people are working together to accomplish a specific task — are one of the primary examples of inverse proportionality. Let's see the same example again, and this time find the correct answer.

✓ Example 2.1.2 Revisited

Suppose it takes 6 sanitation workers, all working simultaneously, 4 hours to pick up the trash and recycling in a given neighborhood. How many sanitation workers would it take to pick up the trash and recycling in the same neighborhood in 3 hours? (Assume that all sanitation workers work at the same rate, and can work independently.)

Solution

The way to approach this is to find the number of *worker hours* needed to accomplish the task of picking up the trash and recycling in this neighborhood. A worker hour is defined to be an hour of work done by a worker, and that number will remain constant no matter the number of workers used.

To find the number of worker hours needed for this particular neighborhood, we simply multiply the known number of workers by the known number of hours:

$$\begin{array}{ccccc} 6 & \times & 4 & = & 24 \\ \text{workers} & & \text{hours} & & \text{worker hours} \end{array} \quad (2.1.15)$$

That means that it will require 24 *worker hours* to pick up the trash and recycling in this neighborhood.

To find the number of workers needed to pick up the trash and recycling in 3 hours, we divide the number of worker hours by the number of hours to find the number of workers:

$$\frac{24 \text{ worker hours}}{3 \text{ hours}} = \frac{24}{\text{worker hours}} \div \frac{3}{\text{hours}} = \frac{8}{\text{workers}} \quad (2.1.16)$$

This means it will take 8 workers 3 hours to pick up the trash and recycling. This makes sense, since it's larger than 6, which was the number of workers needed to accomplish the task in 4 hours.

All inverse proportionality problems work this way — multiply the two known corresponding quantities, and then divide to find the answer. As always, label your units, and check to see if your answers make sense!

2.1.3: Exercises

Make sure that when you are asked to *interpret* something, you write a complete sentence describing the meaning of your numerical answer in the context of the problem.

1. Your car uses 10 gallons of gas to go 300 miles.
 - a. Is this situation described by direct or inverse proportionality, and why? Give a one-sentence answer.
 - b. How many gallons of gas will you need to go 400 miles? Round to the nearest tenth.
 - c. What is the constant of proportionality in this situation, and how would you interpret it?
2. It takes 2 math professors a total of 6 hours to grade the exams for a large math class. Assume all professors grade at the same rate.
 - a. Is this situation described by direct or inverse proportionality, and why? Give a one-sentence answer.
 - b. How many professors would it take to grade the same exams in 4 hours?
3. A family drinks 2 gallons of milk every 9 days. How many gallons of milk will they use in 2 weeks? Be careful with units here! (Round to one decimal place.)
4. At a rate of 30 miles per hour, a certain trip takes 2 hours. How long would the same trip take at 40 miles per hour? (Round to one decimal place or give a fractional answer.)
5. A group of 10 musicians can play a song in 6 minutes. How long would it take for a group of 20 musicians to play the same song? [Hint: Evaluate whether your answer makes sense!!!]
6. Think of a real-world example of direct proportionality that is different than ones we've covered in this section. Give a 2-3 sentence description of the quantities involved, and why you think they are directly proportional. If you use a source, please cite it by providing a URL.
7. Think of a real-world example of inverse proportionality that is different than ones we've covered in this section. Give a 2-3 sentence description of the quantities involved, and why you think they are inversely proportional. If you use a source, please cite it by providing a URL.

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2.2: Rates

We've now studied ratios and proportions, and we've seen that proportions give us a way to solve problems involving ratios. We turn now to a specific type of ratio, called a *rate*, and investigate some interesting applications of rates.

 In this section, you will learn to:

- Recognize and compute rates and unit rates in context
- Use currency exchange rates to calculate the relative values of different types of currency
- Calculate cost per unit in context, and use cost per unit to determine which of two purchasing options is a better deal

2.2.1: Introduction to Rates

We'll start with a simple example that might be relevant to your daily life.

✓ Example 2.2.1

Let's say that your favorite brand of shampoo is Herbal Essences. Bi-Mart sells 11.7 ounces of [this shampoo](#) for \$5.39. On the other hand, you can get 29.2 ounces of this shampoo for \$7.97 [on Amazon](#). What shampoo would give you the better deal?

Sure, the 29.2 ounce shampoo in the above example is more expensive than the 11.7 ounce bottle, but does that mean that the more expensive option gives you less value? How can you compare these two quantities when they have different prices and different volumes? Rates can help us answer this question.

Definition: Rate

A **rate** is a ratio in which the units of the two quantities being compared are different.

In fact, many of the ratios we have seen already are rates. For example, when we computed our hourly pay, we were actually finding a rate, because we compared dollars to hours worked. Other examples of rates include:

- Miles per gallon
- Rotations per minute
- Kilometers per hour

Do you see a pattern with all of these rates? They all have one word in common: the word *per*. This word, which is often ignored, means "for each" or "for every one" in this context. If you see the word "per" between two units, there's a good chance you are dealing with a rate.

In fact, most of the rates we see in our every day lives are a particular type of rate.

Definition: Unit Rate

A **unit rate** is a rate expressed so that the denominator of the corresponding ratio is 1. (Recall that the "denominator" is the bottom number of the ratio when it is expressed in fraction form.)

Think about the "miles per gallon" example above. If we hear, "this car gets 30 miles per gallon," we know that for every 30 miles driven, the car uses 1 gallon of gas. The number 1 is implicit in the way we express most rates. Even if we are given a rate that is not a unit rate, we can convert it to a unit rate using proportions, as in the example below.

✓ Example 2.2.2

Which of the following are unit rates? For those that are not, convert them to unit rates

- 5 miles per hour
- 7 feet every 2 seconds

Solution

- The expression "5 miles per hour" is already a unit rate, since it means "5 miles for every one hour." We can express this ratio in the following way:

$$\frac{5 \text{ miles}}{1 \text{ hour}} \quad (2.2.1)$$

The denominator of this fraction is 1, so this is a unit rate.

- The expression "7 feet every 2 seconds" is not a unit rate. As a fraction, it is represented as follows:

$$\frac{7 \text{ feet}}{2 \text{ seconds}} \quad (2.2.2)$$

The fraction above does not have a 1 in the denominator, so it is not a unit rate. To convert it into a unit rate, we will set up a proportion involving the ratio above, and an equivalent ratio that *does* have a 1 in the denominator:

$$\frac{7 \text{ feet}}{2 \text{ seconds}} = \frac{x \text{ feet}}{1 \text{ seconds}} \quad (2.2.3)$$

$$\frac{7 \text{ feet}}{2 \text{ seconds}} \quad (2.2.4)$$

Now we'll use Cross Multiplication and Division undoes Multiplication to solve for x :

$$\begin{aligned} 2x &= 7 \\ x &= \frac{7}{2} \\ x &= 3.5 \end{aligned}$$

That means that a unit rate equivalent to "7 feet every 2 seconds" is "3.5 feet every one second." This can be rephrased as "3.5 feet per second," which is more easily understood.

An alternative way to solve the second example would simply to divide $7 \div 2 = 3.5$. The fact that these alternative ways give the same answer serves to illustrate how useful our method of solving proportions is in different contexts.

2.2.2: Currency Exchange Rates

We'll start by discussing a bit of background about world currencies and how they are used.

There are over 160 national currencies in the world. When you travel between two countries that use different currencies, it is necessary to exchange some of one type of currency for the other. This is particularly true in countries where cash is used more frequently than credit/debit cards. It's always a good idea when traveling internationally to carry a small amount of that country's currency in either coins or bills. You can get currency from other countries by ordering it through a bank ahead of your travel. You can also exchange currency at a currency exchange business in the airport, port, or border crossing when you are traveling, though these methods usually have fees associated with them and can be exploitative. In some places, ATMs can be used to dispense currency as well.

But there is a big question here: how much of the other type of currency will you get when you exchange? Most countries do not use the United States Dollar or an equivalent, so this question can be hard to answer. For example, Mexico uses Mexican Pesos as its national currency. At the time this book was written, 1 Mexican Peso was worth about 6 US cents according to a [reliable currency exchange website](#). That doesn't mean that product cost significantly more or less in either place. The costs of items are scaled to reflect the fact that one Mexican Peso has a relatively smaller value than one US Dollar. For example, a taco plate in Mexico might cost 90 Mexican pesos, which is equivalent to 15 US Dollars. When the individual units of currency have relatively smaller value, the sticker price on the item goes up. The opposite is also true -- if a currency has a higher value, the items priced in that currency tend to have lower prices. This can be very confusing for travelers.

What's more, these exchange rates change from day to day, depending on international economic factors. At any time, it is possible to look up the exchange rate using the internet, or by going through a bank and asking what exchange rate they are using. There

may even be slight discrepancies between these numbers. In this text, we'll use exchange rates that are current at the time of writing according to Google's currency exchange rate lookup feature.

Since currency exchange rates are, at their most basic mathematical level, ratios, we can use our techniques from earlier chapters to solve currency exchange problems. In this section we will avoid using the \$ sign and instead write out "USD" for US Dollar amounts. This helps avoid confusion with symbols, since other countries also use the \$ symbol to mean the currency of that country.

Let's look at a simple example. Please note that example values in this section may not be current as this book is only updated periodically.

✓ Example 2.2.3

Each Japanese Yen is worth .0093 USD. Prior to a trip to Japan, you want to exchange 250 US Dollars for Yen. How many Japanese Yen will you get in this exchange? Round to the nearest whole.

Solution

We are given the exchange rate of 1 Japanese Yen to .0093 US Dollars. Before we start, let's think for a second about this rate: it means that each Yen is equivalent to less than 1 US cent, because 1 US cent is .01 US dollars, and .0093 is just slightly less than .01. In forming this observation, we can conclude that 250 USD will give us a very large amount of Yen, since each USD will be equivalent to over 100 Yen.

To actually solve this problem, we set up a proportion that compares Yen to USD, using x to stand for the quantity we want to find:

$$\frac{1 \text{ Yen}}{.0093 \text{ USD}} = \frac{x \text{ Yen}}{250 \text{ USD}} \quad (2.2.5)$$

Notice that we've got the exchange rate we are given on the left, and on the right we've got what we're trying to find. The units are labeled so that we know everything is in the correct place. Now we simply solve this proportion:

$$\begin{aligned} \frac{1}{.0093} &= \frac{x}{250} \\ .0093x &= 250 \\ x &= \frac{250}{.0093} = 26,881.72 \end{aligned}$$

Rounding to the nearest whole, we find that we get

$$26,882 \quad (2.2.6)$$

Japanese Yen. This may seem like an enormous amount, but based on our observation before starting this problem, it makes sense that the number of Yen is large.

Currency exchange rate problems can always be solved using a method similar to the one above. Depending on the information you are given, your unknown may occur in a different place, but our techniques for solving proportions allow us to find any unknown value, so long as we have all of the other information.

2.2.3: What's the Better Deal?

To introduce this last application of rates, we'll start with an example. Note that throughout this section, we will use the \$ sign again, and all prices are assumed to be in US Dollars. The next example should look familiar to you.

✓ Example 2.2.4

Let's say that your favorite brand of shampoo is Herbal Essences. Bi-Mart sells 11.7 ounces of this shampoo for \$5.39. On the other hand, you can get 29.2 ounces of this shampoo for \$7.97 on Amazon. What shampoo would give you the better deal?

Solution

Before attempting to solve this problem, let's understand what is being asked. For our purposes, we consider a choice to be the "best deal" if *the cost per unit of the product is lower*. In this case, the unit that is used to measure the shampoo is ounces. Therefore, this question could be rephrased as: which of these two purchasing options has the lowest cost per ounce?

In order to solve this question, we want to determine the cost per ounce of each possible purchase. In other words, we want to find the *unit rate of cost* for each purchase, and the lower unit rate will correspond to the better deal.

For the 11.7 ounce shampoo you can buy at Bi-Mart, we can calculate the cost per ounce by dividing the cost by the number of ounces:

$$\frac{\$5.39}{11.7 \text{ ounces}} \approx \$0.46 \text{ per ounce} \quad (2.2.7)$$

For the \$7.97 ounce shampoo available on Amazon, we calculate the cost per ounce in the same way:

$$\frac{\$7.97}{29.2 \text{ ounces}} \approx \$0.27 \text{ per ounce} \quad (2.2.8)$$

In summary, the 11.7 ounce bottle costs \$0.46 per ounce, and the 29.2 ounce bottle cost \$0.27 per ounce. That means that the 29.2 ounce bottle is a better deal, since it costs less per ounce.

You may have approached the problem above in a different way, by noticing that if you bought two of the 11.7 ounce bottles, you'd only have 23.4 ounces of shampoo, but have paid $2 \times \$5.39 = \10.78 whereas you could have paid \$7.97 for more shampoo, so Amazon is a better deal. That would be a reasonable way to approach this problem, but not all situations are as easy to calculate in your head.

These sorts of questions can get trickier if the units involved are not immediately obvious. Let's see another example.

✓ Example 2.2.5

A pizza parlor sells circular pizzas. A large pizza costs \$12, and a family size pizza costs \$16. A large pizza has a 12 inch diameter, and a family size pizza has a 14 inch diameter. Which pizza is the better deal?

Solution

This problem looks almost too easy at first — certainly the 12-inch pizza is better because it costs a dollar per inch, and the other one costs more per inch!

But wait: what is an "inch" of pizza? That's not a relevant measurement here. The more important unit to consider, when thinking about pizza, is the number of *square inches* of pizza you get. The area of the pizza is what matters most, not the diameter.

Our first step will therefore be to calculate the areas of the pizza involved. In order to do that, you need to know the area of a circle. The area A of a circle with radius r is given by

$$A = \pi r^2 \quad (2.2.9)$$

where π is a special constant number related to circles. While π is irrational, meaning a decimal that goes on forever without repeating, it is usually safe to use $\pi \approx 3.14$ in calculations.

So we proceed to calculate the areas of the two pizzas. The large pizza has a 12 inch diameter. The diameter is twice the radius, so this pizza has a 6 inch radius. We can find area of the large pizza by calculating

$$A_{\text{large}} = \pi r^2 = \pi \cdot (6^2) = 36\pi \approx 113 \text{ square inches} \quad (2.2.10)$$

Similarly, the diameter of the family-size pizza is 14 inches, so its radius is 7 inches. Therefore, the area of the family-size pizza is given by

$$A_{\text{family}} = \pi r^2 = \pi \cdot (7^2) = 49\pi \approx 154 \text{ square inches} \quad (2.2.11)$$

Now that we've got the areas of both pizzas, we can ask: what is the cost per square inch of each pizza? The pizza with the lower cost per square inch will be a better deal.

We find the cost per square inch by dividing the cost of the pizza by the number of square inches. For the large pizza, the cost per square inch is

$$\frac{\$12}{113 \text{ square inches}} = \$0.106 \text{ per square inch} \quad (2.2.12)$$

For the family size pizza, the cost per square inch is

$$\frac{\$16}{154 \text{ square inches}} = \$0.104 \text{ per square inch} \quad (2.2.13)$$

Comparing these two numbers, we see that the \$16 pizza, which was the 14 inch family size pizza, has a slightly lower cost per square inch. Therefore, the family size 14inch pizza is the better deal.

This example illustrates two things:

- It's important to think critically about the question to find the right interpretation. In this case we needed to first find the area of each pizza to make sense of the problem.
- Sometimes cost per unit amounts can be very close, and it's necessary to include many digits following the decimal point. Here it was necessary to go to the third decimal place to see which number was larger.

Keep these things in mind as you answer questions that ask: what is the better deal?

2.2.4: Exercises

1. Write down at least three other examples of rates that were not discussed in this particular chapter.
2. Is 8 meters every 5 seconds a unit rate? If not, convert it to a unit rate.
3. Is 6 beats per minute a unit rate? If not, convert it to a unit rate.
4. Suppose that the exchange rate for US Dollars to Moroccan Dirhams is 1 to 9.67.
 - a. Before a trip to Morocco you exchange 300 US Dollars for Moroccan Dirhams. How many Dirhams do you get?
 - b. At the end of the trip, you have 45 Moroccan Dirhams left. If you exchange these for US Dollars, how many Dollars will you get back?
5. You are told that 1 Swiss Franc is worth 1.04 US Dollars, and that 1 Russian Ruble is worth .014 US Dollars. On a trip from Switzerland to Russia, you exchange 50 Swiss Francs for Russian Rubles. How many Rubles do you get? Round to the nearest whole. [Hint: you can do two separate currency exchange calculations.]
6. Suppose that 32 ounces of soda costs \$1.69, and 54 ounces of the same soda costs \$3.47. Which is the better deal and why?
7. You are purchasing land on which to build a home. There are two rectangular lots to choose from. Lot A is 80 feet by 110 feet, and costs \$94, 000 Lot B is 70 feet by 120 feet, and costs \$87, 000 Assuming that all other characteristics of the properties are equally desirable, and you are willing to accept either size lot, which is the better deal and why? [Hint: a rectangle's area is calculated by its length times its width.]

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2.3: Percents

The final two sections of this part will focus on a very specific type of ratio: a *percent*. In this chapter we'll focus on the one basic equation that describes all problems involving basic percents, and practice using it in context.

 In this section, you will learn to:

- Understand and use the word "percent" in context
- Recognize and convert between various representations of percents
- Use the Basic Percent Equation to solve problems involving percents

2.3.1: Definition and Representation of Percents

We start with the main definition:

Definition: Percent

A **percent** is a ratio with a denominator of 100. The percent sign % is equivalent to the fraction $\frac{n}{100}$.

Recall that the denominator is the bottom of the fraction; that is, the second number in the ratio. The word *percent* literally means "per 100." That is, a percent expressed as a portion of a whole when that whole is divided into hundredths.

A quick note about language: some people use the words *percent* and *percentage* interchangeably, but technically they don't mean the same thing. A percentage is a relative amount (for example, "a large percentage") and a percent is a specific amount (for example, "20 percent"). This isn't a point we'll belabor, but this book will specifically use the word *percent* when referring to a specific ratio.

Before we solve problems involving percents, it's important to know that a percent can be expressed in three ways: using a percent sign, as a decimal, or as a fraction. It is helpful to know how to convert between these. We'll do a couple examples in detail, and then provide a table of other examples for reference.

✓ Example 2.3.1

- Express 96% as a decimal and as a fraction.
- Express $\frac{3}{20}$ as a decimal and as a percent.

Solution

- The percent denoted 96% is equivalent to the fraction $\frac{96}{100}$ by definition of percent. The equivalent decimal is .96.
- Calculating $\frac{3}{20} = 3 \div 20$, we see that the fraction $\frac{3}{20}$ is equivalent to the decimal .15. This decimal is 15 hundredths, or $\frac{15}{100}$ so the corresponding percent is 15%. Note in this example we see that multiple fractions can correspond to the same percent.

In the previous example, you may have noticed a quick way to convert between decimals and percents:

Converting Decimals and Percents

To convert a decimal to a percent, move the decimal point two places *right*, and then put a percent sign at the end. For example:

$$.127 = 12.7\% \quad (2.3.1)$$

To convert a percent to a decimal, move the decimal point two places *left*, and then drop the percent sign. For example:

$$56.3\% = .563 \quad (2.3.2)$$

Additional examples are shown below to make these rules more clear:

Percent	Decimal	Fraction
14%	.14	$\frac{14}{100}$
8%	.08	$\frac{8}{100}$
.5% or "half a percent"	.005	$\frac{.5}{100}$ or $\frac{5}{1000}$
25%	.25	$\frac{25}{100}$ or $\frac{1}{4}$
123%	1.23	$\frac{123}{100}$

2.3.2: The Basic Percent Equation

There is one equation that governs all percent relationships.

Definition: Basic Percent Equation

The equation that describes all relationships involving percents is

$$\text{percent} \times \text{whole} = \text{part} \quad (2.3.3)$$

We will call this the **Basic Percent Equation**. It will be used to solve all basic percent problems.

The Basic Percent Equation can be rearranged using Division Undoes Multiplication in the following way:

$$\text{percent} = \frac{\text{part}}{\text{whole}} \quad (2.3.4)$$

or as

$$\text{whole} = \frac{\text{part}}{\text{percent}} \quad (2.3.5)$$

There is one important caveat to using this equation:

Caution!

To use the Basic Percent Equation, your percent must be expressed as a decimal.

Let's see some examples of the Basic Percent Equation in action.

✓ Example 2.3.2

91% of people in the world are right-handed. In a randomly selected group of 745 people, how many do you expect to be right-handed?

Solution

We need to identify the percent, whole, and part in this equation. The percent, if given, will always have a percent sign next to it. So the percent in this case is 91%, which we need to express *as the decimal*, .91.

In this problem, the number 745 represents the whole, because that is the total number of people. We are being asked to find the *part* of that whole that is right-handed. So we will be finding the part.

We use the Basic Percent Equation:

$$\text{percent} \times \text{whole} = \text{part} \quad (2.3.6)$$

and fill in what we know, which is percent = .91 and whole = 745. We now have

$$.91 \times 745 = \text{part} \quad (2.3.7)$$

Once we compute the left side, we find that

$$677.95 = \text{part} \quad (2.3.8)$$

We will round this to the nearest whole since we're talking about a number of people. Thus, 678 people will be right-handed.

Here's another example:

✓ Example 2.3.3

According to the [WOU website](#), 34% of WOU undergraduates are men. Suppose there are currently 1123 undergraduate men at WOU. How many total undergraduates are there at WOU?

Solution

Once again, we start by identifying the percent, whole, and part in this equation. The percent is 34%, which we express as the decimal .34. In this case, we are being asked to find the total, so the "whole" is the unknown quantity. We are told that the part — the number of undergraduate men — is equal to 1123.

Using the Basic Percent Equation,

$$\text{percent} \times \text{whole} = \text{part} \quad (2.3.9)$$

we substitute what we are given:

$$\text{percent} \times \text{whole} = \text{part} \quad (2.3.10)$$

$$.34 \times \text{whole} = 1123 \quad (2.3.11)$$

Now, since Division undoes Multiplication, we have that

$$\text{whole} = \frac{1123}{.34} \approx 3306 \quad (2.3.12)$$

Since this is a number of people, we rounded to the nearest whole. This means that there are approximately 3306 WOU undergraduates.

In the next section, we'll see some particular applications of percents.

2.3.3: Exercises

Remember to read carefully and answer the question that is being asked!

1. Fill in the missing spots in this table. Copy the entire table in your answer.

Percent	Decimal	Fraction
	.24	
.1%		
	.098	
		$\frac{19}{100}$
238%		
		$\frac{1}{5}$

2. Twelve percent of Polk County residents speak Spanish fluently. There are 9048 fluent Spanish speakers in Polk County. How many total residents are there in Polk County?

3. In a certain dorm on campus, 13 people are social science majors, 12 people are natural science majors, 17 people are education majors, and 9 people have other majors. What percentage of people in the dorm are natural science majors? Round to the nearest tenth of a percent.
4. In a certain acre of forest, there are 457 deciduous trees and 1035 trees in total. What percentage of the trees in this acre of forest are non-deciduous? Round to the nearest tenth of a percent.
5. You're trying to save up to put a 10% down payment on a house. You hope to purchase a \$315,000 house. Your plan to save equal amounts of money each month for four years to reach your goal. How much will you need to save each month? (Assume the saved money earns no interest.)
6. Read [this article](#). After reading the article, answer the following questions:
 - a. Oregon's population in 2020, when this article was written, was estimated to be 4,455,920 people. According to the statistics given in this article, what number of people in Oregon are non-Hispanic white people? Make sure to show your calculation.
 - b. Assume there are 135,438 people of color under the age of 15 living in Oregon. According to a statistic given in this article (in the second half), calculate how many total people under the age of 15 live in Oregon. Make sure to show your calculation.
 - c. Answer at least one of the following questions, writing at least 2 sentences.
 - Did anything surprise you when reading this article? If so, what was it?
 - Are you curious about other statistics relating to the population of Oregon? If so, what would you ask?

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2.4: Applications of Percents

We've now learned about several ways to describe numbers and their relationships: ratios, rates, proportions, and percents. In this section, we'll explore more applications of these concepts to real-life situations.

 In this section, you will learn to:

- Accurately compute marked down prices using percents
- Accurately compute and interpret the percent change between two quantities

2.4.1: Computing Markdowns

In this section, we will learn how to compute prices that have been *marked down* by a percent of the original prices. A markdown is a form of discount in which a percent of the original price is subtracted. This is a frequent way in which sale prices in stores are expressed — as a "percent off" the original price. We'll use one main formula:

$$\text{Marked Down Price} = \text{Original Price} - (\text{Markdown Percent} \times \text{Original Price}) \quad (2.4.1)$$

In the equation above, the Markdown Percent must be expressed as a decimal! That's because the original price is the "whole" and the markdown percent is the "percent," so we are finding the "part" that represents the portion of the original price being subtracted from the original price. In other words, the "part" is the amount a consumer would save as compared to the original price. An example should make this clear:

✓ Example 2.4.1

The original price of a pair of pants is \$24.99. There is a 30% off sale at the store where the pants are being sold. If you buy these pants, how much will you pay for them? How much do you save as compared to the original price? Round to the nearest cent.

Solution

This is a straightforward use of the markdown equation, shown above. The original price is \$24.99, and the mark down percent is 30%. As a decimal, this is 0.30, because we move the decimal point two places to the left. Therefore, our mark down equation gives us

$$\text{Marked Down Price} = 24.99 - (0.30 \times 24.99) \quad (2.4.2)$$

Now, we need to be careful with the order of these operations. The multiplication inside the parentheses happens first, so we can compute $0.30 \times \$24.99 = \7.497 . Therefore, we have

$$\text{Marked Down Price} = 24.99 - 7.497 = 17.493 \quad (2.4.3)$$

The instruction is to round to the nearest cent. Recall that cents consist of two places to the right of the decimal point. Therefore, when we round to the cents place, we get that the marked down price is \$17.49, which is what we will pay for these pants during the 30% off sale. Since $\$24.99 - \$17.49 = \$7.50$, we are saving \$7.50 as compared to the original price of the pants.

This is the essence of a mark down — it takes a specific percentage of the price and subtracts it from the original price to give the discounted price.

You may have noticed a slightly more efficient way to accomplish this. Instead of using the original markdown equation as it was written, we could reason as follows: If the pants are marked down 30%, that means they cost 70% of their original price, since $100\% - 30\% = 70\%$. Thus, we could simply compute

$$0.70 \times 24.99 = 17.493 \quad (2.4.4)$$

and after rounding we get that the sale price of the pants is \$17.49, matching the answer we have in the solution above. As usual, either way of answering the question is fine — you should pick the method that makes the most sense to you. However, this second

way of thinking about the problem, in which we subtracted the markdown from 100%, will be helpful in the next example, so it is worth making sure you understand both.

We can ask another type of question with markdowns: what if we want to find the original price given the marked down price? The following example illustrates how to do this.

✓ Example 2.4.2

A coat was marked down by 25%. Its cost after the markdown is \$41.24. What was the original price of the coat? Round to the nearest cent.

Solution

This is a bit trickier to figure out — our formula above does not easily lend itself to solving this problem. Instead of using the markdown formula, we'll instead use what we know about percents.

The key intuition for this problem is: if the coat was marked down by 25%, its new cost is 75% of the original cost, since $75\% = 100\% - 25\%$. Since the marked down price is calculated by subtracting a percent of the original cost, the original cost can be viewed as 100%, and the cost after a markdown of $n\%$ is $(100 - n)\%$ of the original cost.

In our case, this means that the new price of the coat, \$41.24, is 75% of the original cost of the coat. Therefore, we can use the Basic Percent Equation with $\text{percent} = 0.75$ and $\text{part} = \$41.24$. We will solve for "whole" in this equation, and that answer will be the original price of the coat.

Therefore we have:

$$.75 \times \text{whole} = 41.24 \quad (2.4.5)$$

Now we'll use Division undoes Multiplication to find the whole:

$$\text{whole} = \frac{41.24}{.75} = 41.24 \div .75 = 54.99 \quad (2.4.6)$$

where we've already rounded to the nearest cent. Therefore, the original cost of the coat was \$54.99.

Next time you're at the store, see if you can find mark downs phrased this way. You can test your understanding by computing your savings!

2.4.2: Percent Change

It's common to see phrases like the "the population of North Plains, Oregon increased by 58% from 2016 to 2021." In this case, there are percents being used, but it's not exactly stating that one number is a certain percent of another number. Instead, it is describing a percent by which one quantity changed with respect to its original value. While the previous formulas involving percents can be used to understand problems like this, it's sometimes easier to reframe them slightly into the following formula, specifically designed to deal with situations where a percent change is being described.

Definition: Percent Change

Given two quantities Q_1 and Q_2 , the **percent change** between the quantities is

$$\text{percent change} = \frac{Q_2 - Q_1}{Q_1} \times 100 \quad (2.4.7)$$

Note that in this definition, we are multiplying by 100, so there is no need to convert to a percent afterwards. The answer will automatically be a percent.

Typically, when using this formula, the two quantities involved are ordered. Q_1 represents the "first value," and Q_2 represents the "second value." As usual, it will be necessary to read carefully and think critically to determine what the values of Q_1 and Q_2 are in context. Here is an example to see how this works:

✓ Example 2.4.3

According to climate.gov, between the years 1990 and 2000, there were 52 weather-related natural disasters in the United States that cost more than 1 billion dollars to mitigate. Between 2010 and 2020, there were 119 weather-related natural disasters in the United States that cost more than 1 billion dollars to mitigate. Comparing these two decades, what is the percent change in weather-related natural disasters from the first decade to the second? Round to the nearest tenth of a percent.

Solution

There are a lot of numbers in this question! Let's focus on the important ones. We see that during the earlier decade, from 1990 to 2000, we have 52 disasters. That is our first quantity, since it occurs at an earlier time. That means $Q_1 = 52$. Likewise, in the second decade from 2010 to 2020, we have 119 disasters, so $Q_2 = 119$. The years are simply telling us what happens first and second; they do not factor into the equation at all, and neither does the 1 billion dollars — that is just part of the description of the facts we are looking at.

Using $Q_1 = 52$ and $Q_2 = 119$, we have

$$\text{percent change} = \frac{Q_2 - Q_1}{Q_1} \times 100 = \frac{119 - 52}{52} \times 100 \quad (2.4.8)$$

Then

$$\text{percent change} = \frac{67}{52} \times 100 \quad (2.4.9)$$

When working with division and multiplication together, we work left to right, first computing the fraction as a decimal, and then multiplying by 100:

$$\text{percent change} \approx 1.288461 \times 100 = 128.8461 \quad (2.4.10)$$

The instructions were to round to the nearest tenth of a percent, so we will state the answer as 128.5%. This means that the number of natural disasters costing more than 1 billion dollars rose 128.8% between the decade from 1990 – 2000 to the decade from 2010 – 2020

It is important to note what happens if the two quantities, Q_1 and Q_2 , represent a decrease, meaning that Q_2 is smaller than Q_1 . In this case, the percent change formula will output a negative number. This simply means that there is a decrease by the given percent. This does not indicate a wrong answer! It is important to make sure that whatever Q_1 is, it's in the denominator of the fraction. That's why it's always important to label your variables and rewrite the original equation before proceeding.

2.4.3: Exercises

In this section, when there is a percent as an answer, please round to the nearest percent. If your answer is a number of people, round to the nearest whole. And if your answer is an amount of money, round to the nearest cent.

1. A dresser has an original price of \$120, but it is marked down 25%. If you buy the dresser with this markdown, how much will you pay for the dresser?
2. A TV originally cost \$350. It was marked down 15% when you went to buy it. You also have a coupon that takes 10% off your total purchase. When you buy it, how much money do you save off of the original price?
3. A pair of shoes currently has a price of \$31.44. You see a sticker that says the current price is a 15% mark down off of the original price. What was the original price of the shoes?
4. The price of Google stock on April 16th, 2020 was \$1266.78. The price of Google stock on April 17th, 2020 was \$1257.43. What is the percent change of the Google stock price between these two days? Round to the nearest hundredth of a percent.
5. This question has two parts. Remember to check to that your answers make sense to you.
 - a. The population of Oregon was 12,093 in 1850 and 90,923 in 1870. What is the percentage change in the population between those two years? Round to the nearest percent.
 - b. The population of Valsetz, Oregon was 300 in 1983 and 5 in 1984 (it is now a ghost town — this is what the dining hall at WOU is named for). What is the percentage change in population between those two years? Round to the nearest percent.

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CHAPTER OVERVIEW

3: The Language of Lines

See how a common type of relationship between numbers can be used to solve problems and make predictions.

[3.1: Functions](#)

[3.2: Linear Functions](#)

[3.3: Linear Modeling with Data](#)

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3.1: Functions

In this section, we will study one of the most fundamental mathematical objects — the function. Later in the book, we will use particular types of functions to describe real-world relationships and make predictions using data. However, we need to start with some general knowledge and concepts involving the notion of a function.

 In this section, you will learn to:

- Recognize examples and non-examples of functions represented in various forms
- Read and compute with functions represented in various forms
- Use function notation to describe the inputs and outputs of various functions in context

3.1.1: Introduction to Functions

 Definition: Function

A **function** is a relationship between a set of inputs and a set of outputs such that each input has exactly one output.

Of course, there are many other non-mathematical meanings of the word *function*. For example, *function* often means the same thing as *purpose* or *capacity*, but that's not quite what the word means mathematically. A more similar meaning is found in the sentence, "a child's height is a function of their age." What this sentence means is that "a child's height tends to depend upon their age." This is similar to our meaning of the word function, but it's still not quite the same. The mathematical word *function* indicates a dependence — the outputs will usually be dependent upon the inputs — but we also require that each input correspond with exactly one output.

Functions are useful in a number of real-world contexts. Scientific trials are often designed to detect the effect of an input variable on an output variable. As an example, consider a clinical trial of a pharmaceutical drug. A doctor might want to measure the percent of the drug remaining in a person's body at various times after the person has ingested the drug. That relationship is *functional* in the sense that, at each particular amount of time following the ingestion, there is a particular amount of the drug left. A doctor could look at the data of the times and percents, begin to see a pattern, and make predictions based on those observations. The fact that each time value is related to a single percent value means this relationship is a function.

We will explore several examples and non-examples of functions so you can see what sorts of relationships qualify as functions. We will also learn several different ways to express functions, each of which will be useful in different contexts.

We start with functions described in words. We use an example involving Social Security numbers. We acknowledge that not every person reading this book will have a Social Security number; however, it is a useful example to explain this concept. The background information that you need is: every citizen of the United States is assigned a 9-digit number, called their Social Security number, that is unique to them during their lifetime (meaning they are the only living person who is associated to that number). It is used as an identifying number when filing government documents. For the purposes of this example, we will ignore the possibility of identity theft.

✓ Example 3.1.1

- Consider the relationship whose input is a person's Social Security number and whose output is their name. Is this relationship a function? Why or why not?
- Consider the relationship whose input is a person's name and whose output is their Social Security number. Is this relationship a function? Why or why not?

Solution

- This relationship is a function. The inputs are Social Security numbers and the outputs are names. For each input, which is a Social Security number, there is exactly one name associated to that number. Said another way, a single number cannot be associated to two separate names. Therefore, each input has exactly one output, and so this relationship is a function.

- This relationship is not a function. It is certainly the case that some names are associated with multiple Social Security numbers. For example, the name Leanne Merrill has at least three Social Security numbers associated to it. Therefore, the input "Leanne Merrill" has at least three different outputs. Therefore, this relationship is not a function, because there is *more than one* output associated to one of its inputs.

This example shows that a relationship can be a function "in one direction" and not in the other. That is, when we took the Social Security numbers as inputs and names as outputs, we had a function. But when we went the other way, from names to Social Security numbers, it was no longer a functional relationship. This is important to keep in mind!

3.1.2: Function Representations

Now that we've defined the notion of a function, we will talk about different ways to represent functions. These representations will help us understand why functions are useful in different contexts.

We will look at four different ways to represent functions:

- Using words
- Using a table
- Using a formula
- Using a graph

We have already seen how to express functions in words. The Social Security example from the previous section shows how a function might be expressed in words.

Next, we'll look at functions expressed via tables, often called a *table of values*. When reading a function as a table, read across: the output associated with a given input is in the same row.

✓ Example 3.1.2

Do the following tables represent functions? Why or why not?

1.	Input	Output
	0	1
	1	2
	2	3
	3	1

2.	Input	Output
	1	3
	2	5
	4	9
	2	1

Solution

In this example, Table 1 is a function and Table 2 is not.

In Table 1, each input is associated to a single output. There is no input that has multiple outputs. Therefore, Table 1 represents a function.

In Table 2, the input 2 has two different outputs, namely 5 and 1. Therefore, Table 2 does not represent a function, since there is an input that has more than one output.

A common issue that occurs when learning about functions is to confuse the notions of *function* and *pattern*. Looking at Table 1 from the previous example: we said that Table 1 represents a function. But you may have noticed that the outputs do not appear to follow a predictable pattern. If there were a pattern, we might expect that the output of 3 is 4. But it's not — it's 1. Additionally, it may seem like a problem that the *output* of 1 is repeated. That's also allowed in a function! The thing to avoid is repeated *inputs*.

Of course, many functions do follow patterns, and some functions have unique outputs as well. But those are not strictly necessary for the definition of function.

Next, we'll look at functions expressed using formulas and graphs. We'll see that tables play an important role.

✓ Example 3.1.3

Consider the function $y = -3x + 10$, where x is the input and y is the output. Make a partial table of values for y using the values $x = 0, 1, 2, 3$. Then sketch a graph of the function $y = -3x + 10$.

Solution

In this example, we will use the given formula to find a table of values for this function. We will take the values of x that are given and substitute them, or "plug them in," to the formula for the function. When $x = 0$, we have

$$y = -3(0) + 10 = 10 \quad (3.1.1)$$

Therefore, if the input of this function is 0, the output is 10.

Likewise, if $x = 1$, we have

$$y = -3(1) + 10 = 7 \quad (3.1.2)$$

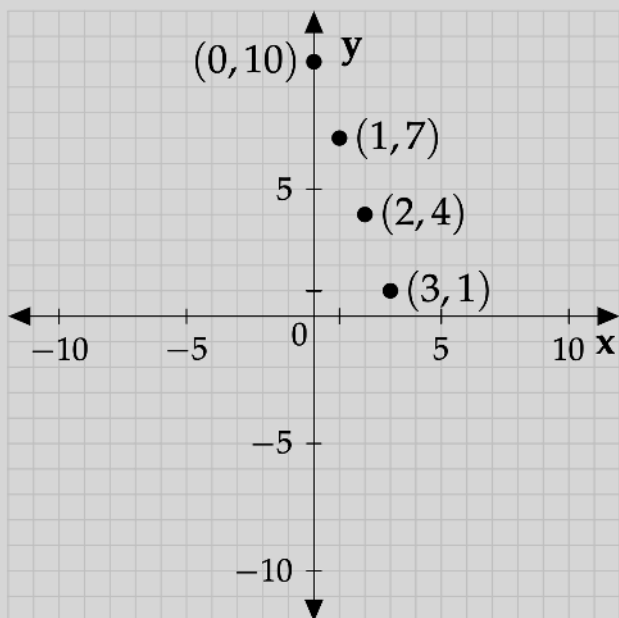
Therefore, if the input of this function is 1, the output is 7.

We continue in this way. We see that the input 2 has output 4, and the input 3 has output 1.

We will summarize this information using a *table of values* for the function:

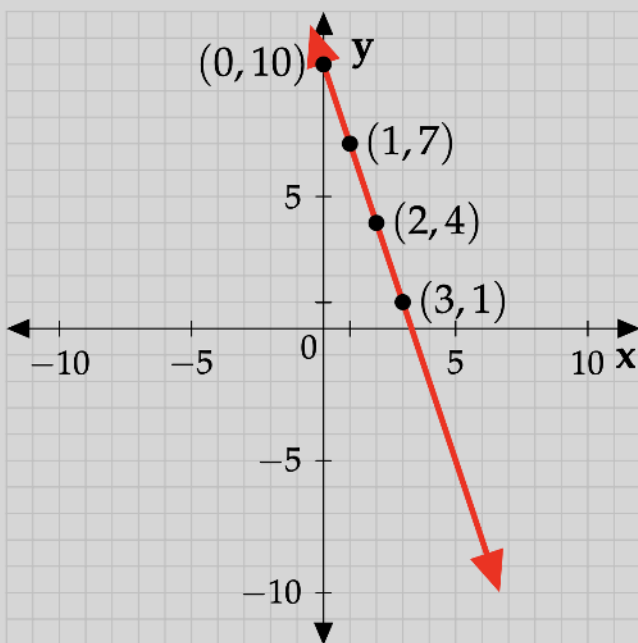
Input	Output
0	10
1	7
2	4
3	1

Now, we will draw the graph for this function. In order to do that, we will view the rows of the table as *ordered pairs* (x, y) . We will show these ordered pairs on a graph.



As a quick review of graphing: the first number x in the ordered pair (x, y) denotes the location of the point relative to the horizontal axis (which is often labeled the x -axis, as it is here). The second number y in the ordered pair (x, y) denotes the location of the point relative to the vertical axis (which is often labeled the y -axis, as it is here). So, for example, the point $(3, 1)$ lies at a value of 3 on the horizontal axis and a value of 1 on the vertical axis. This point is uniquely described by this ordered pair.

Now that we have the points from the table, we will sketch the smoothest curve that we can between these points. In this case, the "curve" is a straight line. Therefore, the graph of this function will look like the thick line in the picture below:



The technique outlined in the previous example is very useful. If we are given a function as a formula, we can create a table of values for that function, and then use that table of values to graph the function. This technique ties together the formula, table, and

graph representations of a function.

3.1.3: Function Notation

When we work with functions, it is helpful to have the following notational convention.

Note

We will often give a function a single letter name, such as f . We write $f(x)$ to mean "the output of f associated to x ." This symbol $f(x)$ is said verbally as " f of x ."

The parentheses denote the place where the input is "plugged in" to the function f . If we choose a value to substitute for x , we say that we are *evaluating* f at that value of x .

We highlight a common confusion about this notation below.

Caution!

The notation above, $f(x)$, does NOT indicate that f is being multiplied by x . It symbolizes f as a function with an input of x .

Let's see an example to understand this notation.

✓ Example 3.1.4

Define a function f by $f(x) = 2x^2 + 1$. Find the following values:

- a. $f(0)$
- b. $f(3)$
- c. $f(-1)$

Solution

We will find these values using the definition of function notation. For $f(0)$, we will express

$$f(0) = 2(0)^2 + 1 = 2 \times 0 + 1 = 0 + 1 = 1 \quad (3.1.3)$$

That is, we are "plugging in," or substituting, the number 0 where x is. Then we simplify to get a numerical answer. This is the general way that *function evaluation* works.

For $f(3)$, we will use a similar technique. We will substitute $x = 3$ into $f(x)$:

$$f(x) = 2(3)^2 + 1 = 2 \times 9 + 1 = 19 \quad (3.1.4)$$

So that means $f(3) = 19$.

For $f(-1)$, we need to be careful with order of operations, remembering that exponents happen before multiplication. We have:

$$f(-1) = 2(-1)^2 + 1 = 2 \times 1 + 1 = 3 \quad (3.1.5)$$

as a negative multiplied by a negative becomes positive. Therefore, $f(-1) = 3$.

We will end with an example of function notation in context. Note that will use different letters for the function and its input here. We often do this if the context calls for it. Example 3.1.5

✓ Example 3.1.5

Let $C(g) = 10g + 60$ be the function that gives the cost of a monthly cell phone plan, in dollars, based on the number g of gigabytes of data used by the cell phone. Find the amount that the plan will cost if 4 gigabytes of data are used.

Solution

We have that $C(g)$ is the cost associated to g gigabytes of data used. That is, the input is the number of gigabytes of data used, and the output is the cost of the cell phone plan in dollars. If you use 4 gigabytes of data, that means that $g = 4$. That means the cost of the cell phone plan will be $C(4)$, which we can calculate as

$$C(4) = 10(4) + 60 = 10 \times 4 + 60 = 40 + 60 = 100 \quad (3.1.6)$$

Therefore, if 4 gigabytes of data are used, it will cost \$100.

We will see soon how useful functions can be in making predictions and summarizing data.

3.1.4: Exercises

Throughout this homework, if you need to make a graph you can either hand-write or use an online graphing program (such as [Desmos.com](https://www.desmos.com)) to create a graph.

1. Give an example of a function described in words that is different from those given in the text. (Something similar to the Social Security example, but a different context). You should write 2-3 sentences to explain your example.
2. Consider the following tables. Which ones represent functions, and which ones do not? Explain your reasoning for each table in a sentence or two.

a.

Input	Output
3	0
-5	-1
0	8
3	1

b.

Input	Output
1	42
2	42
3	42
4	42

3. Find the following quantities. Please round decimal answers to three places.
 - a. Let $f(x) = 5x - 4$. Find $f(1)$.
 - b. Let $g(x) = 3x^2 + 9$. Find $g(0)$.
 - c. Let $h(x) = \frac{5x-1}{7x+2}$. Find $h(2)$.
 - d. Let $l(x) = .01(2^x)$. Find $l(6)$.
4. Suppose that the price P of a gallon of gas, in dollars, is a function of the demand d of gas, in billions of gallons used in a given day. The function is given by $P(d) = 1.34d$.
 - a. Evaluate $P(2)$ and interpret your answer. That is, say what your numerical answer means in the context of the problem.
 - b. Make a table of values for $P(d)$, using the values $d = 0, 1, 2, 3$
 - c. Graph the function $P(d)$ based on the table of values from the previous part. You can do this by hand or use a graphing program such as [Desmos](https://www.desmos.com).
 - d. Respond to the following prompts in 2-4 sentences: Does this seem like an accurate model of the price of gas? Why or why not? What additional restrictions or caveats might you suggest to make this model more accurate?

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3.2: Linear Functions

In this section, we will focus on a particular type of function known as a *linear function*. Much can be said about linear functions, and in fact there is an entire branch of mathematics, *linear algebra*, devoted to the study of linear functions! We will focus more on the applications of linear functions in this chapter.

 In this section, you will learn to:

- Recognize examples and non-examples of linear functions
- Use linear functions to model real-world situations
- Describe various geometric properties of linear functions, including their slope

3.2.1: Linear Functions

Instead of starting this section with a definition, as usual, we will start with an example.

✓ Example 3.2.1

For most residential customers who are connected to city water, the [city of Monmouth](#) charges a basic monthly water use \$22.48 fee each month, plus a \$.0323 per cubic foot of water used. Write a function that describes the cost of a monthly water bill in Monmouth in terms of the number of cubic feet of water used. (Note: these numbers are accurate as of 2023.)

Solution

Before we start trying to write a function, let's think about a few examples. If you used 100 cubic feet of water (a very small amount for monthly water use), you would have to pay $100 \times \$0.0323 = \3.23 for that 100 cubic feet of water, plus the basic fee charge. So the total would be calculated as follows:

$$\left(\underset{\text{rate per cubic foot} \times \text{cubic feet}}{\$0.0323 \times 100} \right) + \underset{\text{basic fee}}{\$22.48} = \$25.71 \quad (3.2.1)$$

That is not too bad, since we are using exactly 100 cubic feet. What if you use 200 cubic feet? Well, in that case, we have to pay the volume rate twice. In that case, we would have:

$$\left(\underset{\text{rate per cubic foot} \times \text{cubic feet}}{\$0.0323 \times 200} \right) + \underset{\text{basic fee}}{\$22.48} = \$28.94 \quad (3.2.2)$$

That is, we are taking the number of cubic feet of water, multiplying it by the rate per cubic foot, and then adding the basic fee. Either way, the basic fee does not change from month to month, but the amount you pay for water does -- it is determined by your usage.

So, for a general formula, we could use w to stand for the number of cubic feet of water used, and $C(w)$ to stand for the cost of the water bill in dollars. Remember: this is function notation, and it simply means that the cost is dependent upon the amount of water used. It is not denoting multiplication!

Therefore, a function that represents this situation is:

$$C(w) = 0.0323w + 22.48 \quad (3.2.3)$$

That may not have been obvious, but now that you've seen it, perhaps you can recognize similar situations in the future. There are two components to this situation: a flat fee, and a per unit rate. The per unit rate is multiplied by the number of units used. In the previous example, the units were cubic feet, and the per unit rate was \$.0323 per cubic foot of water. The basic fee, or flat fee, is a cost that does not change based on usage, and is just added onto the bill. In the previous example, the flat fee was \$22.48. There are many things that are charged this way. We will see some examples, and if you think about bills you or your family pay, you can likely think of more.

This common function structure, which shows up not only in bills but in many other places, has a special name.

 Definition: Linear Functions

A **linear function** is a function that can be written $f(x) = mx + b$ for some numbers m and b . The number m is called the **slope** of the function, and represents the rate of change of the function. The number b is called the **vertical intercept** of the function, and represents the starting value of the function. The graph of a linear function looks like a straight line.

Let's see a couple examples.

 Example 3.2.2

Is the water bill example from the previous problem a linear function? If so, find its slope and vertical intercept and interpret them in the context of the question. Evaluate $C(500)$, graph the corresponding point, and explain its significance.

Solution

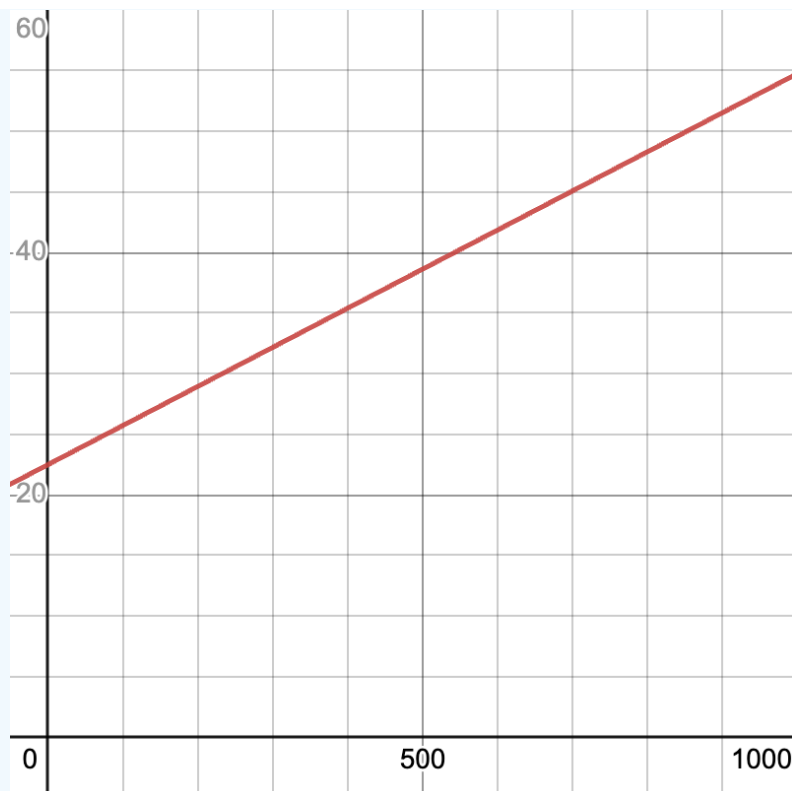
The formula from the previous problem is indeed a linear function. Recall, we found that:

$$C(w) = 0.0323w + 22.48 \quad (3.2.4)$$

where w was the number of cubic feet of water used and $C(w)$ was the cost associated to that usage. We see that indeed, this has the form $mx + b$, where $m = 0.0323$ and $b = 22.48$. You may notice that instead of the variable x and function name f , the function $C(w)$ uses a C and a w instead. This is intentional, and shows that the letter names for the variable and function don't affect whether the function is linear. We are free to choose letters as we see fit to match the context of the problem. In this problem, w stood for the amount of water, and C stood for the cost of the bill, which will help us remember their meaning when we need to interpret our answer later.

Since we know $C(w) = 0.0323w + 22.48$ is linear, we can see that its slope is 0.0323 and its vertical intercept is 22.48. In context, the slope represents the fact that each cubic foot of water costs \$0.0323 that is, it is the *rate of change* of the cost with respect to usage. The vertical intercept in this case is 22.48, which represents the flat fee, or starting value. In other words, it is the amount you would pay if you used no water at all.

To graph this, we can use the methods we learned in the previous section: either we can make a table of values, or we can use graphing software (such as [Desmos](#)). Desmos produces this picture:



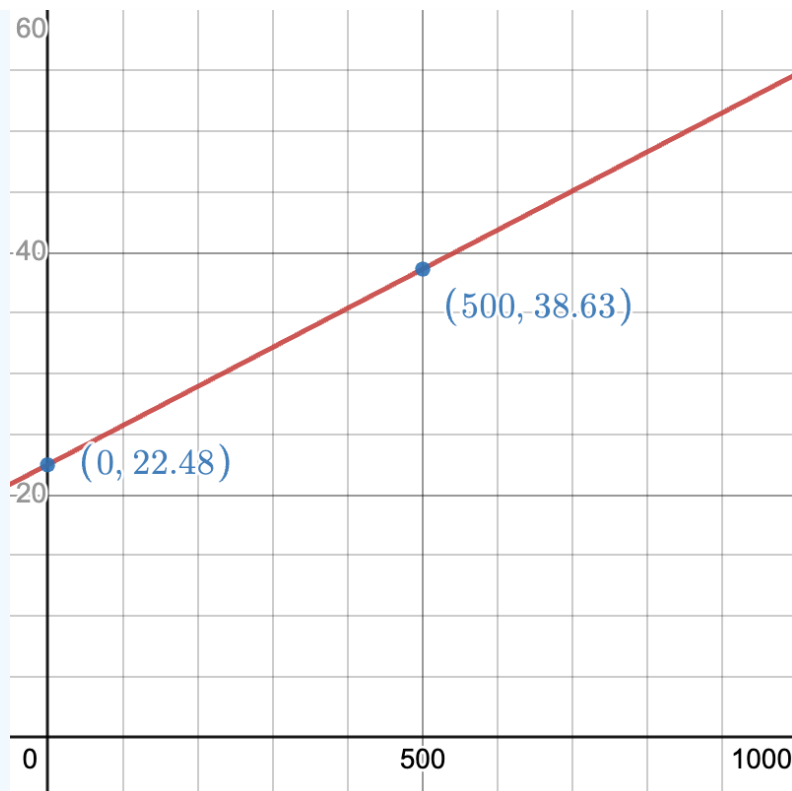
On this picture, we see that the line crosses the vertical axis just above the 20 line. The exact spot it is crossing is at 22.48, which is the vertical intercept of the function. That shows the flat fee that is paid when 0 cubic feet of water are used.

As we move to the right, the horizontal axis shows how many cubic feet of water are used. As that number increases, the value of the function (which is the vertical value of the red line) increases. For example, when the number of cubic feet of water used is 500, we can look at 500 on the horizontal axis. The value of the function at that point, which is between 35 and 40 on the vertical axis, is the cost associated with 500 cubic feet of water usage. We can calculate the exact value of this point by computing:

$$C(500) = 0.0323 \times 500 + 22.48 = 38.63 \quad (3.2.5)$$

That is, if you use 500 cubic feet of water, your water bill will be \$38.63. The graph gives us an easy way to estimate any the cost associated with any number of cubic feet. The formula allows us to calculate exact values.

Here is the graph again, this time with the relevant points labeled.



Take a look at the calculations again, and make sure that you understand how they relate to the graph. The first coordinate of each ordered pair corresponds to the amount of water used, and the second coordinate corresponds to the cost of the bill. The graph gives us a way to visualize the entire relationship between water used and cost.

3.2.2: Properties of Linear Functions

Something you may have noticed about the water bill example is that the cost went up at a constant rate when compared to the water usage. This is the hallmark of linear functions -- a constant rate of change. This constant rate of change can either be increasing or decreasing. If it is an increasing linear function, the same amount will be added to the output for every increase of one in the input. If it is a decreasing linear function, the same amount will be subtracted from the output for each increase in the input.

✓ Example 3.2.3

Each of the following tables represents a function. Which one(s) could be linear? Explain why you know.

a.

x	$f(x)$
1	2
2	4
3	8
4	16

b.

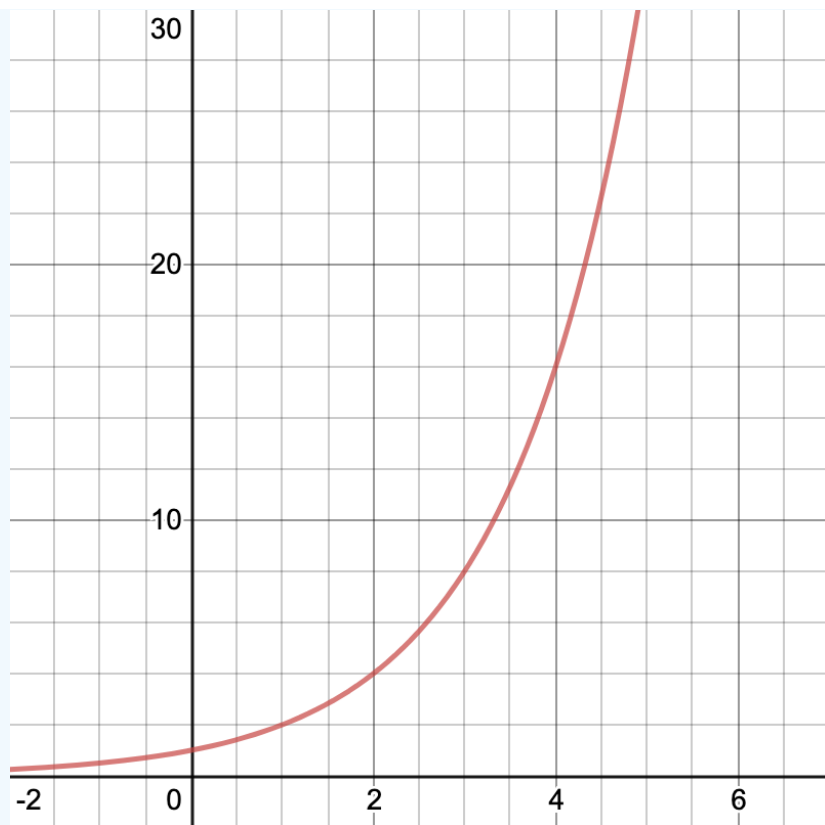
x	$g(x)$
1	4
2	1
3	-2
4	-5

Solution

Let's look at the table a, which describes a function, $f(x)$ first. We see that the inputs (that is, the x values) are increasing steadily by 1. If this were a linear function, we should be able to add the same amount to each successive output (that is, the $f(x)$ values) to predict the behavior of the function. However, we see that the distance from $f(1) = 2$ to $f(2) = 4$ is 2 but the distance from $f(2) = 4$ to $f(3) = 8$ is 4. That is, the outputs are increasing at different amounts, rather than the same amount. Therefore, $f(x)$ is not a linear function.

Now, you may correctly notice that there is a predictable pattern to function $f(x)$. The outputs here are *multiplied* by a constant number each time. Even though this is a pattern that we can recognize mathematically, $f(x)$ still does not qualify as a linear function. It does not have a *constant rate of change* -- meaning the same amount added each time -- which is necessary for a linear function.

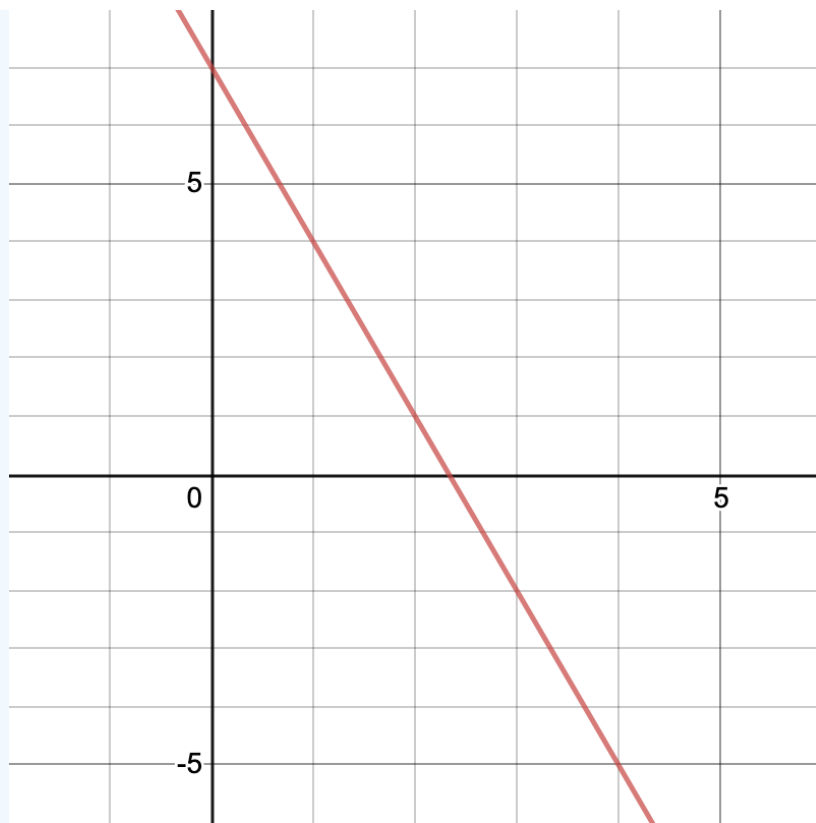
If you are still not convinced that this isn't a linear function, examine its graph, which you can find by hand if you plot points and connect them using a smooth curve:



Notice that this graph is not a straight line; rather, it is a curve that has an increasing rate of change. That is, as the input (horizontal) values increase, the output (vertical) values increase at a faster and faster rate. This means the function $f(x)$ cannot be linear, because its graph is not a straight line.

Now, we move onto the table b, which describes a function $g(x)$. We notice that the inputs of $g(x)$ change by 1, so the outputs should either increase or decrease at a constant rate. Indeed, as we move from $g(1) = 4$ to $g(2) = 1$, we decrease by 3. Then from $g(2) = 1$ to $g(3) = -2$, we decrease by 3 again. Likewise, from $g(3) = -2$ to $g(4) = -5$ is a decrease by 3.

Therefore, table b could represent a linear function. If we plot the points and connect them, we get the following graph:



This graph is a straight line, which is further geometric confirmation that $g(x)$ could be a linear function.

You may wonder how to take the data given in the previous example and make a linear function from it. There is a process to do this, which involves a slightly more careful look at the concept of *slope*.

Definition: Slope

The **slope** of a linear function is the rate of change of the linear function. That is, the slope is the constant amount of increase or decrease for each change of 1 to the input of the function. It is calculated as

$$m = \text{slope} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1} \quad (3.2.6)$$

where (x_1, y_1) , (x_2, y_2) are any two points on the line.

Here is a brief explanation of the formula: we are comparing two points on the line, (x_1, x_2) and (y_1, y_2) , and calculating the ratio of "change in y " to "change in x ." To do that, we calculate $y_2 - y_1$, which is the vertical distance between the two points, also known as the "rise." Likewise, the quantity $x_2 - x_1$ is horizontal distance between the two points, also known as the "run". The slope is the ratio of these two quantities, read as "rise over run." It describes the change in the output y as it relates to the change in the input x . An example will help illustrate this.

✓ Example 3.2.4

According to [Honda's website](#), a new basic model 2013 Honda accord cost \$21,680 when it was first sold. In 2023, such a car should sell for about \$13,086 according to [Kelly Blue Book](#). Assuming that the price of a 2013 Honda Accord changes linearly over time:

- Find the slope of the linear function describing the price of a Honda Accord, and interpret it in context.
- Find a formula for the linear function that describes the price of a Honda Accord t years after 2013 and interpret the vertical intercept in context.

- Use your function to predict the price of a Honda Accord sedan in 2033, and label the corresponding point on the graph.
- Find the year in which the price of a 2013 Honda Accord is projected to be \$7929.60

Solution

First, please take note that this is an oversimplified situation. Used car prices, while they do tend to decrease over time, are affected by market factors that make their value highly variable and subject to fluctuation. This exercise shows how to use linear functions to get a ballpark idea, but this should not be mistaken for an exact prediction!

Our first step is to take our two data points and convert them into ordered pairs. Here, a convenient input variable will be the time t in years since 2013, and the output variable will be the price $P(t)$ in dollars. Our first point will be $(0, 21680)$ which corresponds to the fact that 0 years after the year 2013, a Honda Accord cost \$21,680. Our second point will be $(10, 13086)$ which corresponds to the fact that 10 years after 2013, that same Honda Accord should \$13,086. This process of converting the written information into mathematical points is always the first step of the mathematical problem-solving process.

Now we can use these points to stand in for (x_1, y_1) and (x_2, y_2) in the slope formula. We have:

$$(x_1, y_1) = (0, 21680) \quad \text{and} \quad (x_2, y_2) = (10, 13086) \quad (3.2.7)$$

Next, we'll use the slope formula to find the slope:

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{13086 - 21680}{10 - 0} = \frac{-8594}{10} = -859.4 \quad (3.2.8)$$

Therefore, the slope of the linear function is -859.4 . Since the slope was calculated by dividing a number of dollars by a number of years, its units are dollars per year. That is, the slope is $-\$859.40$ per year. This means that a 2013 Honda Accord loses value at a rate of \$859.40 per year. (Such a value loss is known as *depreciation*.)

To answer the second part, we need to work backwards a bit. We observe that the vertical intercept will correspond to the year $t = 0$, which in this case is the year 2013. We have that the value of the car in 2013 is \$21,680. Therefore, our vertical intercept is 21680.

Now that we have our slope and vertical intercept, we can write the equation of the linear function describing the cost of a 2013 Honda Accord t years after 2013:

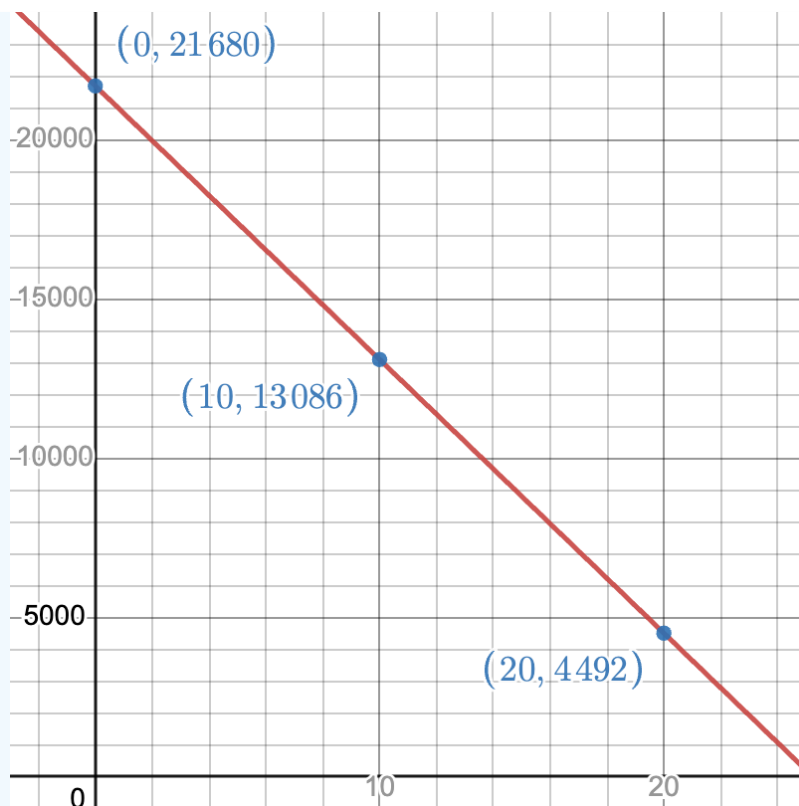
$$C(t) = -859.4t + 21680 \quad (3.2.9)$$

To answer the last part, we will note that 2033 is 20 years after 2013, and therefore it corresponds to the value $t = 20$. Thus, we simply need to evaluate the function $C(t)$ at the value $t = 20$. We have:

$$C(20) = -859.4 \times 20 + 21680 = 4492 \quad (3.2.10)$$

That is, in the year 2033, a 2013 Honda Accord is predicted to be worth about \$4,492.

We can visualize the line and the values at all three points in time by graphing the line and labeling the points:



Our last question asks us to find the year in which the projected price of a 2013 Honda Accord is equal to \$7929.60. This is slightly different from the previous question in that, instead of knowing the year and finding the price, we are going backwards — we know the price, and we are trying to find the year in which that price occurred. That means that instead of plugging in a value for t , we are instead plugging in a value for $C(t)$ and then solving for t . This will require two steps of algebra, but one of them is already familiar to you from a previous chapter.

We'll start by plugging in our cost, \$7929.60 for $C(t)$:

$$7929.60 = -859.4t + 21680 \quad (3.2.11)$$

Our goal is to isolate t . In order to do that, we need to use the fact that *subtraction undoes addition*. In other words, we will subtract 21680 from both sides:

$$7929.60 - 21680 = -859.4t + 21680 - 21680 \quad (3.2.12)$$

On the right side, we see that $21680 - 21680 = 0$ so we are left with

$$7929.60 - 21680 = -859.4t \quad (3.2.13)$$

We can then perform the subtraction on the left side, which gives us a negative number:

$$-13750.4 = -859.4t \quad (3.2.14)$$

Now we have a situation that may look familiar from the previous chapter: we can use *division undoes multiplication*! We get

$$t = \frac{-13750.4}{-859.4} = 16 \quad (3.2.15)$$

This tells us that when $t = 16$, the price is equal to \$7929.60. Rephrasing the answer in context, we see that in 2029 (which is 16 years after 2013), the price of a 2013 Honda Accord is \$7929.60. See if you can find the corresponding point on the graph above!

This shows the importance of linear functions in making predictions about the future. Of course, not all relationships are linear. But often linear functions give close-enough estimates for many situations, and are relatively simple to calculate. We'll see more of how

linear functions are used in making predictions in the next section.

3.2.3: Exercises

- Go to [Monmouth Power and Light's website](#). Find the Rates link on the side menu to look for the current Residential electricity rates. Locate the basic customer charge (flat fee) and the cost per kilowatt hour (kWh). (Be careful not to round here, and note that some of the numbers are given in cents but others are given in dollars.)
 - Write a linear function $C(k)$ that describes the cost of your Monmouth electric bill in dollars, in terms of the number of kilowatt hours k that are used.
 - [According to the EPA](#), The average American household uses 886 kilowatt hours of energy each month. Find the cost of your electric bill in Monmouth if you use 886 kilowatt hours of energy in a given month using the function $C(k)$.
 - Graph the function $C(k)$ either by hand or using an electronic graphing tool such as [Desmos](#). Label the point corresponding to your answer from the previous question on the graph.
- Of the following three tables, one does not represent a function, one represents a function that is not linear, and one represents a linear function. For each part, identify which property the table satisfies. Justify your answer in each part using a complete sentence.

a.

Input	Output
1	4
2	16
3	64
4	256

b.

Input	Output
1	7.5
2	5
3	2.5
4	0

c.

Input	Output
1	3
2	5
3	7
1	9

- A population of deer in a forest is 87 in 2015, and 175 in 2023. Assuming that the deer population changes linearly:
 - Find the linear function $P(t)$ that describes the population of deer in the forest t years after 2015.
 - Using your function from the previous part, predict the deer population in 2030. (Round to the nearest deer if applicable.)
 - Find the year in which the deer population is projected to be 417 deer.
 - Graph your function either by hand or using Desmos, and label the point corresponding to your answer from the previous two parts.
 - What are some limitations to this model? What is unrealistic about it?

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3.3: Linear Modeling with Data

Linear functions have many uses in other areas of mathematics. For instance, the subject of calculus is based on using linear functions to approximate more complicated functions, and linear functions in multiple dimensions are used to describe physics and chemistry applications. In this section, we will see how linear functions are used in an important statistics context: linear regression.

 In this section, you will learn to:

- Recognize, interpret, and make a two-variable scatter plot using given data
- Find the average of a set of numbers
- Perform a linear regression based on a set of two-variable data
- Find and interpret the coefficient of correlation based on a set of two-variable data

3.3.1: Scatter Plots and Definitions

Much like the last section, we'll start with an example to make sense of the concepts we'll see in this section.

✓ Example 3.3.1

3.14 *a la mode*, a fictitious ice cream parlor in Seattle, wishes to better predict its monthly sales numbers. It recorded the average monthly high temperature each month, along with the monthly sales in thousands of dollars, for a year, and compared them using the table below.

Month	Avg. High Temp (Deg. Fahrenheit)	Sales in \$1000 s
January	47	18
February	50	30
March	54	32
April	58	65
May	65	54
June	70	61
July	78	66
August	76	77
September	71	75
October	60	45
November	51	39
December	48	18

Analyze the data above in the following ways:

- Find the average monthly temperature, in degrees Fahrenheit, throughout the year.
- Find the average monthly ice cream sales, in thousands of dollars, throughout the year.

Then, from this table, plot the numerical values on each row as (x, y) coordinates, and describe trends that you see in the graph. Can you explain these trends?

Solution

- a. We haven't seen averages yet in this book, but you may be familiar with the concept from previous mathematical experiences. An *average*, or *mean*, of a list of numbers is found by adding up all the numbers, and then dividing by how many numbers were in the list. The average is known as a "measure of center" of a set of data, which can sometimes help to summarize a key fact about the data. However, it doesn't tell us about the individual data points themselves, or how spread out they are. Nevertheless, let's find the average of the monthly temperatures. We first add up the numbers:

$$47 + 50 + 54 + 58 + 65 + 70 + 78 + 76 + 71 + 60 + 51 + 48 = 728 \quad (3.3.1)$$

Then, we'll divide by how many numbers were in the list; in this case, there are 12 numbers (corresponding to the 12 months of the year):

$$728 \div 12 = 60.667 \quad (3.3.2)$$

We see that the decimal continues, but we choose to round to the thousandths place. What does this number tell us? It tells us that the average monthly temperature in Seattle for the year above was 60.667 degrees Fahrenheit.

- b. We'll use a similar approach to find the average ice cream sales. First, we'll add up the numbers in that column:

$$18 + 30 + 32 + 65 + 54 + 61 + 66 + 77 + 75 + 45 + 39 + 18 = 580 \quad (3.3.3)$$

Then, as before, we'll divide by 12:

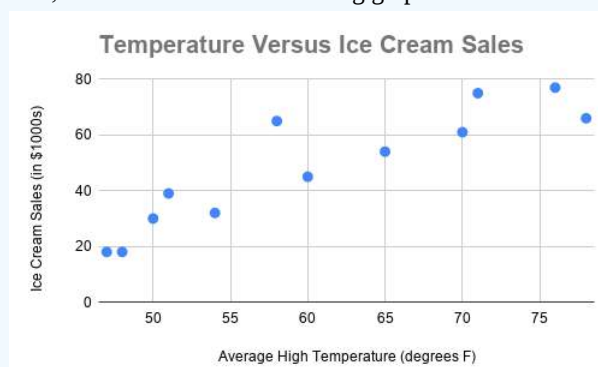
$$580 \div 12 = 48.333 \quad (3.3.4)$$

Once again, we see that the decimal continues, so we will choose to round to the thousandths place. This tells us that the average monthly ice cream sales, in thousands of dollars, are 48.333 thousand dollars per month. A clearer way to say this would be that average monthly ice cream sales are \$48,333 dollars per month.

Next, we'll plot the numerical values of each row as (x, y) coordinates. This means we'll take the numbers in a given row and simply think of them as an ordered pair. For example, in the first row, the numbers are 47 and 18. In that case the ordered pair would be $(47, 18)$. If you'd like to think of 47 as the "input" and 18 as the "output" you may, though beware of that intuition later on, because it won't always hold up. The first column of the table, which lists the months, will actually not be a part of our scatter plot. The months are here simply to organize the data into related pairs. We will not eventually care about months -- we are simply interested in the relationship between temperature and sales.

In any case, we generate our list of ordered pairs: $(47, 18)$, $(50, 30)$, $(54, 32)$, and so on. Then we will plot these points on a graph. I'll show how to do this using a computer in the screen cast for this section, or you can do it by hand.

In any case, when we plot these points, it will look like the following graph:



It's not necessary to label any of the points, but we should make sure that, in general, the graph looks correct before moving on. Indeed, we see two dots in the lower left next to each other. We see that these correspond to the points $(47, 18)$ and $(48, 18)$, which correspond to January and December. You can go through all of the other points and verify that they appear. One thing you will notice when you verify this is that the horizontal axis does not start in a regular way -- it skips from 0 to 50, and then increases in increments of 5. This is a perfectly fine thing to do when making a chart, provided the axes are clearly labeled. The reason we do this is to center our data in the ranges that matter, and in fact most good graphing programs will do this automatically.

There are many possible ways we could describe the trends in the graph. It looks like, as the temperature increases, the ice cream sales also increase. Though, it could also be described as somewhat random. For example, compare the two points in the middle of the graph that appears to go "down" — these points are (58, 65) and (60, 45). This means that when the average high temperature was 58 degrees, there were \$65,000 in sales, but when the temperature rose to 60 degrees, the sales went down to \$45,000. So it's not a perfectly stable relationship.

Also, take note that the average sales and average temperature occur in approximately the "middle" of the graph. This makes sense, as these numbers told us something about the central tendency of our data. However, as we can see from the graph, there is a lot of variability from the average, and that's perfectly normal. This demonstrates that the average can be a useful estimation, but actual numbers will vary in context.

Can we explain these trends? Well, given the data at hand, we can't make any certain conclusions about the relationship between ice cream sales and temperature. However, these data do indicate some shared relationship between temperature and ice cream sales, because both quantities appear to rise together. Thinking about the context of the situation, this makes sense. Using common sense, we can say that people will probably buy more ice cream when it's hotter. The graph of the points gives us a visual way to confirm that supposed relationship.

The graph in the question above is called a *scatter plot*, and the question illustrates how a scatter plot can visually suggest a relationship between two sets of data. However, scatter plots alone have two problems:

- They do not give us the ability to clearly state how strongly related two sets of data are — we can only describe relationships in qualitative terms.
- They do not give us the ability to make a conclusion involving *causality*. That is, we cannot say that one variable necessarily influences or affects the other.

Of course, in a given context, we may have very good reasons to believe that there is a causal relationship between the variables. For instance, in the previous example, it seems quite likely that an increase in temperature would cause ice cream sales to rise. As we will see, we cannot usually make such claims, and sometimes variables that appear very strongly related from a graphical standpoint actually have nothing to do with one another.

Before we work on the second bullet point above, let's try to remedy the first one. We need a way to clearly state how strongly related two sets of data are. This brings us to the following two definitions.

Definition: Line of Best Fit

The **line of best fit** or **trendline** is a linear function that approximates a scatter plot by minimizing the distance between the line and each of the points in the scatter plot. The process of finding the line of best fit is **linear regression**.

Definition: Coefficient of Correlation

The **coefficient of correlation**, commonly referred to as R , is a numerical measure of the amount of correlation between two variables. That is, it measures how closely related the two variables are.

These two definitions will help us more clearly describe our scatter plot. We can find the line of best fit using linear regression, and then compute the value of R to get specific information about how closely related the data are.

Procedure

To learn how to perform a linear regression, watch [this video](#). In doing so, you will find the equation for the line of best fit, and a value called R^2 .

To find R from R^2 , follow these steps:

1. Take the square root of R^2
2. Assign either a positive or negative sign to the result. Assign a positive sign if the trendline slope is positive, and a negative sign if the slope is negative.

Once you find R , consult the following chart to determine the type of correlation that data exhibits.

Range of R value	Interpretation
$-1 \leq R < -.8$	strong negative linear correlation
$-.8 \leq R < -.5$	moderate negative linear correlation
$-.5 \leq R < 0$	weak negative linear correlation
$R = 0$	no linear correlation
$0 < R < .5$	weak positive linear correlation
$.5 \leq R < .8$	moderate positive linear correlation
$.8 \leq R \leq 1$	strong positive linear correlation

In other words, if R is close to 1 or -1, that means that the data are very closely correlated; they move together in lockstep. However, if R is closer to 0, the data are not correlated at all. The table above shows the gradation as R changes values.

However, as usual, we need to use caution when we are interpreting data.

Caution!

Caveat: The ranges for weak, moderate, and strong correlation will vary based on context. The ranges above are what we will use for this class but should not be interpreted as universally applicable.

To expand a bit: different disciplines have different correlation needs and conventions. For example, in a psychology experiment, a value of $R = .4$ might be considered moderate positive correlation. And in biology, a value of $R = -.78$ could be considered weak negative correlation. You will learn the convention of your discipline in relevant discipline-based classes. However, the ranges indicated above are good enough for many situations, and represent a happy medium between many different disciplines.

3.3.2: Regression and Correlation Examples

Let's see an example of regression and correlation in context.

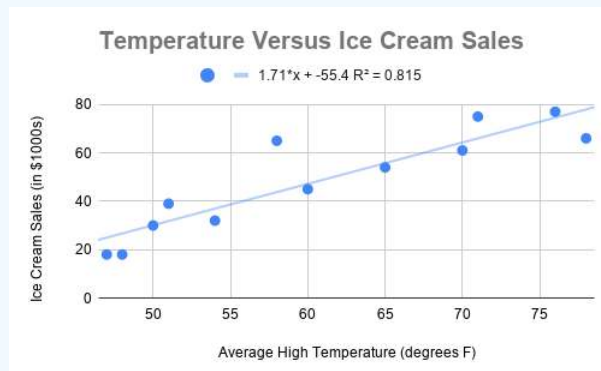
✓ Example 3.3.3

For the ice cream sales data above, perform a linear regression. Find the value of R and state the type of correlation exhibited. Then, use the line of best fit to predict ice cream sales if the temperature is 67 degrees.

Solution

You just saw how to do this in the example video! If you didn't watch it before, [watch it now](#).

Once we have the linear regression, including the value of R^2 , we produce the following graph:



From the top of the graph, we can see that the line of best fit is $S(x) = 1.71x - 55.4$. We are choosing to name our function $S(x)$ since the output of the linear function represents sales. We also see that $R^2 = .815$.

To find the value of R , we will first take the square root of R^2 :

$$\sqrt{R^2} = \sqrt{.815} = .9028 \quad (3.3.5)$$

Now, we ask whether the correlation is positive or negative. Since the slope of the line of best fit is positive (in this case, it's 1.71, the correlation is also positive. Therefore, we have $R = .9028$. Looking at the chart for R values, we see that this data exhibits *strong positive correlation*. This confirms our suspicion from before!

Finally, we will use the line of best fit to make a prediction. The input to this line, x , stands for the temperature. Therefore, to predict sales when the temperature is 67 degrees, we will substitute 67 in for x in the equation for $S(x)$. That is, we have:

$$S(67) = 1.71(67) - 55.4 = 59.17 \quad (3.3.6)$$

Remember that the units on sales were thousands of dollars. So, we predict the sales will be \$59,170 if the temperature is 67 degrees. Looking at the graph, this seems reasonable, since the line of best fit appears to be at about 60 on the vertical axis when the horizontal axis value is 67.

This shows how finding the line of best fit can help us make predictions, and how finding R gives us a more specific and quantitative way to describe the relationship between two sets of data. Unfortunately, it did not tell us anything more about causality. As we will see in the next example, R does not have anything to do with causation.

✓ Example 3.3.4

Below are the data that compare the number of engineering Ph.D.s awarded in a given year to per-person mozzarella cheese consumption in the US. Make a scatter plot of these data, and find the R value. What kind of correlation does this data exhibit?

Year	Engineering Ph.D.s Awarded	Cheese eaten (in pounds)
2000	480	9.3
2001	501	9.7
2002	540	9.7
2003	552	9.7
2004	547	9.9
2005	622	10.2
2006	655	10.5
2007	701	11
2008	712	10.6
2009	708	10.6

Solution

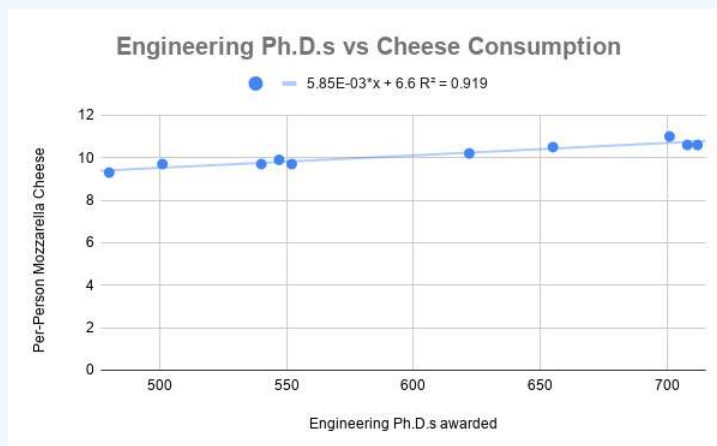
If this question does not seem strange to you, go back and read it again.

Why are we comparing engineering Ph.D.s and per-person mozzarella cheese consumption? These two variables have no obvious reason to be related, and it seems virtually impossible that a change in one variable would cause a change in the other. In other words, awarding more engineering Ph.D.s should not cause cheese consumption to increase (or decrease). Likewise, even if the average person eats more cheese, that should not affect whether more or fewer people receive engineering Ph.Ds.

(Some have suggested that perhaps hungry and cash-strapped graduate students are driving pizza consumption up — while there is some truth to this, the proportion of engineering graduate students to the entire US population is so minuscule that it

would not make that large of a difference.)

In any case, we really shouldn't expect to see any relationship between these two variables, and therefore we may also not expect to see any correlation. Let's make the scatterplot and see:



Amazingly, these data look very related. Look how close all of the points are to the line of best fit! That means that they must be closely correlated. Let's find R to be sure. We calculate:

$$R = \sqrt{R^2} = \sqrt{.919} = .9586 \quad (3.3.7)$$

Since the slope of the trendline is positive (in other words, both the variables are increasing together), we have $R = .9586$. That means that these two sets of data — engineering Ph.D.s and per-person cheese consumption — are strongly positively correlated. Indeed, this R value is above .95, and thus very close to 1, which stands for perfect correlation (all points on the line of best fit). One might even say these are very strongly positively correlated!

The example above is from a website called [Spurious Correlations](#). The main goal of this website is to illustrate an extremely important concept in statistics:

Caution!

Correlation does not imply causation.

In other words, even if two variables are strongly correlated, there may not be a causal relationship between them. That is, the variables may not influence one another at all, despite being mathematically linked.

The reason to emphasize this point is immense. It is very, very easy to forget this point when we read the news or make decisions about how we spend our time, money, and effort. For example, suppose you read the headline: "Higher rates of free/reduced lunch in schools linked to low test scores." In the article, you see that the authors found a strong negative correlation between school district poverty and test scores. From this, it is *very easy to assume* that receiving free/reduced-price lunch *causes* lower test scores. But be careful. The way in which one pays for lunch should not, on the face of it, impact a given test score. There are multiple possible ways in which this could be explained, of course. There are likely third factors (called lurking variables) that would influence *both* free/reduced lunch rates and test scores. Or there could be some other factor -- for example, the presence or lack of a particular test prep program in a school -- that would explain the changes in test scores. Such a discussion requires significantly more careful thought, analysis, and knowledge. The headline does not provide this, and instead suggests a causal link that is likely not there.

3.3.3: Exercises

1. The following fictitious table lists the number of hours per week a high school student uses their cell phone, and compares that

with the student's high school GPA.

Phone Use (Hours Per Week)	High School GPA
2.5	3.8
3	3.9
4	3.5
4	3.5
7	3.6
7.5	3.1
8	2.9
8.5	2.9
10	2.5
11	2.3
14	2.1
14	1.9
14	3.9

- Find the average phone use hours per week for this set of data.
 - Find the average GPA for this set of data.
 - Perform a linear regression on this data, and make a graph that includes the scatter plot, line of best fit, and value of R^2 .
 - Find the value of R (be careful with signs here!) and state which type of correlation this data exhibits.
 - Write a 3-4 sentence answer to the following prompt: Do you believe that there is a causal relationship here? If so, in what direction? If not, what are other explanations for this correlation?
2. Find an example of a news headline from the last 6 months in which a correlation between two quantities is implied. Read the article and determine whether or not a causal relationship existed in this case. Write a 4-5 sentence summary of your findings, and include the title and author of the article, as well as the source (just a URL is fine).

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CHAPTER OVERVIEW

4: Exploring Exponential Equations

Encounter a new type of relationship between numbers and practice using it in real-life situations.

[4.1: Exponential Functions](#)

[4.2: Simple and Compound Interest](#)

[4.3: Annuities](#)

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4.1: Exponential Functions

In this section, we will study a different family of functions, and investigate their role in financial mathematics.

 In this section, you will learn to:

- Recognize examples and non-examples of exponential functions
- Distinguish between exponential growth and exponential decay using context clues and algebra
- Identify initial values and growth rates in context and use them to write exponential functions

4.1.1: Exponential Functions: Basic Intuition and Definition

We will start this section with a hands-on exercise. You are encouraged to actually complete the task before moving on for maximum impact.

✓ Example 4.1.1

Find a regular piece of paper (lined paper, printer paper, used scratch paper are all fine -- but you want approximately 8.5 inches by 11 inches in size). This paper is about 0.1 millimeters thick. Your goal will be to describe the thickness of the stack of paper as you fold it in half multiple times.

To get you started: after 0 folds, the stack of paper is 0.1 millimeters thick since there is only one piece of paper in the stack. Now fold the paper in half. Then there are two sheets of paper in the stack, so the entire stack is 0.2 millimeters thick.

Now fold it in half again, but perpendicular to your first fold. Your paper should look like this:



There are four sheets of paper in this stack, meaning that if you were to draw a line through the center of the stack from top to bottom, you would intersect the paper four times. The thickness of this stack is 0.4 millimeters.

Number of folds	Number of sheets	Thickness (in millimeters)
0	1	0.1
1	2	0.2
2	4	0.4
3		
4		
5		
6		
7		

Here are some questions to think about:

- Can you find a *function* that describes the thickness of the stack based on the number of folds?
- How many folds would it take for the stack of paper to reach the moon? (Pretend your paper is large enough to keep folding it as many times as you need.)

In the previous exercise, you may have noticed several patterns in your table, even if you were not able write a function that described the situation. This is a difficult question because the behavior above is not described by a linear function, which is the only type of function we've studied in detail. Instead, it is a new type of function, called an exponential function.

Definition: Exponential Function

An **exponential function** is a function that can be written $f(x) = a(1 + r)^x$ for some numbers a and r . The number r is called the **growth rate** or **decay rate** of the function, and represents the percent change of the function as a decimal. If r is positive, it is a growth rate, and if r is negative, it is a decay rate. The number a is called the **initial value** of the function, and represents the starting value of a function.

To contrast linear and exponential functions further, we provide the following table:

	Linear Functions	Exponential Functions
Formula	$f(x) = mx + b$	$f(x) = a(1 + r)^x$
Where x is	Multiplied by slope	In the exponent
Initial Value	b	a
Change Factor	m	r
Behavior	Consistent amount of change	Consistent percent change
Graph	Line	Curve

Whereas linear functions increase or decrease by the same *amount* between equally spaced values of x , exponential functions increase or decrease by the same *percent*.

✓ Example 4.1.1 Revised

Let's look again at the paper folding example above. We'll fill in the table and relate it to the definition of exponential function.

Solution

Here is the table with the correct values filled in:

Number of folds	Number of sheets	Thickness (in millimeters)
0	1	0.1
1	2	0.2
2	4	0.4
3	8	0.8
4	16	1.6
5	32	3.2
6	64	6.4
7	128	12.8

You may have noticed a pattern in the Sheets column: the numbers are doubling each time you fold the paper. In other words, they are being multiplied by 2. Yet another way to say this idea is: the number of sheets is increased by 100% each time you fold the paper. This phrasing — increasing by 100% — shows us that this function is exponential.

Now that we know this is an exponential function, we'll set about finding the values a and r in the exponential function:

$$f(x) = a(1 + r)^x \quad (4.1.1)$$

for this particular situation.

We said that the number r in the exponential function $f(x) = a(1 + r)^x$ is the growth rate. In general, this will correspond to the percent change between two equally spaced x values in an exponential function. So in this function we will take $r = 1.00$, which comes from converting 100% to the decimal form.

The initial value a is the starting value for the function. In this case, the starting value is 0.1, since that's the thickness of the stack after 0 folds. So we have $a = 0.1$.

Therefore, our exponential function is:

$$f(x) = 0.1(1 + 1.00)^x \quad (4.1.2)$$

which can be written more succinctly as

$$f(x) = 0.1(2)^x \quad (4.1.3)$$

Here, the variable x stands for the number of folds you've make, and the value $f(x)$ is the corresponding thickness of the stack of paper. In case you are not convinced that this is the correct formula for $f(x)$, let's pick a value of x to try. If we folded the paper 5 times, we would use $x = 5$. Then we would calculate:

$$f(5) = 0.1(2)^5 = 0.1(32) = 3.2 \quad (4.1.4)$$

which is indeed the answer we found on the table above. In general, verifying your answer by testing a point or two in this way is encouraged!

To answer the second question — how many folds will it take to get to the moon — we have to do a bit more work. To answer it directly requires logarithms, which we haven't seen yet. But you could hope to do a guess-and-check approach now that you have the function. You'll also need to know how far away the moon is: 384,400,000 meters. This is equivalent to

384,400,000,000 millimeters. So, we are asking when the output of the function $f(x) = 0.1(2)^x$ will be equal to 384,400,000,000. That is, for what x is

$$0.1(2)^x = 384,400,000,000 ? \quad (4.1.5)$$

Do you have an answer in mind? Many people guess in the hundreds or thousands, if not more. After all, the stack was only 12.8 millimeters thick after 7 folds. So it *must* be a large number!

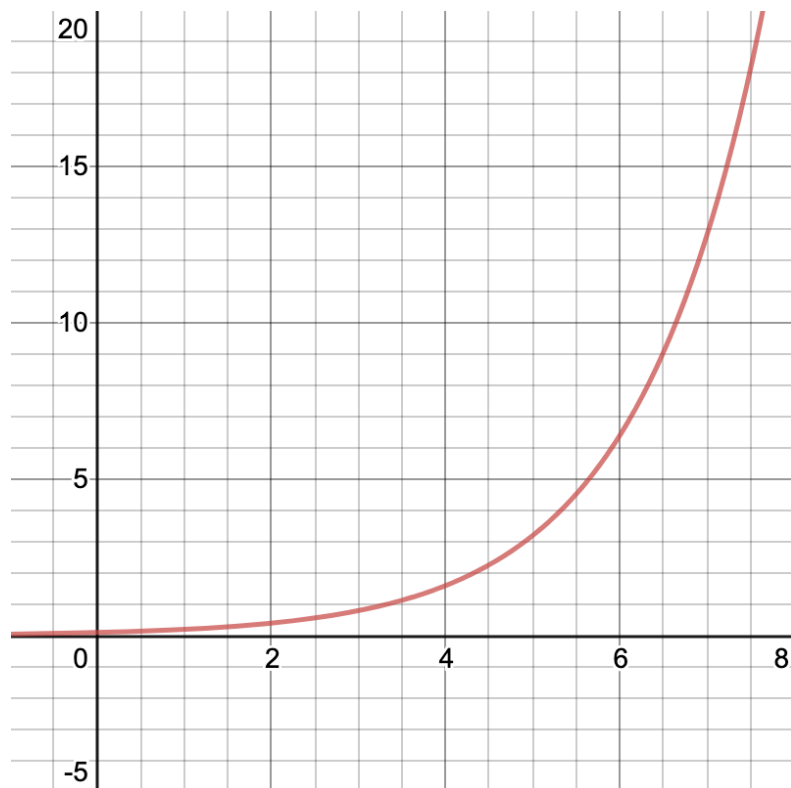
Astoundingly, the answer is 42. Seriously. It turns out that

$$f(42) = 0.1(2)^{42} = 439,804,651,110.4 \text{ millimeters} \quad (4.1.6)$$

which is more than the distance to the moon. (However, 41 folds is not quite large enough, so to reach the moon, we need at least 42.)

The example above is meant to illustrate how extreme exponential growth can be. Of course, it is impossible to actually fold a piece of paper (even a large one) 42 times. It's common to only get 5 or 6 folds on a standard piece of printer paper, and it's a somewhat common myth that it's "impossible" to fold paper more than 7 times. Check out the excellent [Mythbusters segment](#) on this question!

In any case, it should be clear from the example above that exponential growth and linear growth behave very differently. Here is a graph of $f(x) = 0.1(2)^x$ to further illustrate:



This graph is not a line, but rather a curve that slopes upward as it goes. Graphically, this means that exponential growth functions increase at an increasing rate. Linear functions, on the other hand, increase at a constant rate. This is the fundamental difference between linear and exponential functions.

4.1.2: Exponential Functions in Context

Exponential functions are used to describe many different real-world situations. Any situation in which there is a constant percent rate of change can be described using an exponential function. If the function is increasing, the percent rate of change will be positive, and the situation will be modeled using exponential growth. If the function is decreasing, the percent rate of change will be negative, and the situation will be modeled using exponential decay. We will see an example of each scenario.

✓ Example 4.1.2

You are hired at a new job (congratulations!). Your starting salary is \$38,000 and you are promised a raise of 2.5% per year. Write a function $S(t)$ that describes your salary t years after starting the job. How much will you make 4 years after starting the job? How about 5 years after starting the job? And what is the percent change between these values?

Solution

We see that this situation has an initial value and a consistent percent change. Therefore, it must be an exponential function. As in previous sections, we can be flexible about the variable names we use in our functions. This question is asking for a function $S(t)$, so we will use S in place of f and t in place of x in our generic exponential format. This helps us keep track of the fact that our input variable is time t , and our output is $S(t)$, the salary after t years. So, our function will have the format:

$$S(t) = a(1 + r)^t \quad (4.1.7)$$

where a is the initial value and r is the percent rate of change. Since you start with a salary of \$38,000 we have $a = 38000$. Since there is a growth of 2.5% per year, we have $r = .025$. Thus, our exponential function is:

$$S(t) = 38000(1 + .025)^t \quad (4.1.8)$$

We can add the numbers in parentheses to simplify slightly. And in fact, this will help us more easily calculate using this function. That is, we can write

$$S(t) = 38000(1.025)^t \quad (4.1.9)$$

Now, to find your salary 4 years after you start working, simply plug $t = 4$ into the equation:

$$S(4) = 38000(1.025)^4 = 41944.89 \quad (4.1.10)$$

Make sure, when checking the answer above, that you perform the exponent operation before multiplying by the initial value. Therefore, your salary after 4 years will be \$41,944.89. We are rounding to two decimal points because this is an amount of money.

We can use a similar process to find the salary after 5 years. We have:

$$S(5) = 38000(1.025)^5 = 42993.51 \quad (4.1.11)$$

Therefore, your salary after 5 years will be \$42,993.51

To find the percent change between these quantities, we remember from Chapter 2 that the percent change formula is given by

$$\text{percent change} = \frac{Q_2 - Q_1}{Q_1} \times 100 \quad (4.1.12)$$

We'll use $Q_1 = 41,944.89$ and $Q_2 = 42,993.51$. Plugging in, we get

$$\text{percent change} = \frac{Q_2 - Q_1}{Q_1} \times 100 = \frac{42993.51 - 41944.89}{41944.89} \times 100 \approx 2.49999 \quad (4.1.13)$$

Note that this is almost exactly 2.5%. The very small error is due to the fact that we rounded the salaries to two decimal places. This confirms that our constant percent change persists year after year in this function.

The previous example shows a very straightforward application of the exponential function formula. We are presented with a percent change and an initial value, and we can simply write down the function and then use it.

A variation on this theme occurs when an exponential function is *decreasing* rather than increasing.

✓ Example 4.1.3

A new car costs \$35,000 and its value depreciates at a rate of 12% each year. Find the formula $V(t)$ that describes the value of a new car t years after you buy it. Find $V(t)$ and interpret your answer in the context of a problem using a complete sentence.

Solution

The key word to understanding this questions is "depreciates." Something *depreciates* if it loses value over time. Many large purchases, such as cars, boats, and industrial equipment, depreciate over time. In fact, one of the tasks that accountants face is determining how to calculate depreciation for objects over time so they can determine the entire value of a person's or company's possessions at a given time. In this situation we will use a relatively simplistic model of depreciation, in which value decreases by a fixed percentage each year.

Since this is a situation involving a decreasing quantity, we need to choose r to be negative here. The percent change will therefore be -12% , or $-.12$. The initial value remains positive, so we have $a = 35000$. If we call our function $V(t)$ for the value of the car at time t , we have:

$$V(t) = 35000(1 + (-.12))^t \quad (4.1.14)$$

which we can simplify as

$$V(t) = 35000(.88)^t \quad (4.1.15)$$

From this form, we can view this situation slightly differently. The .88 in parentheses indicates that the car *retains* 88% of its value from year to year. This is equivalent to losing 12% of its value. (This is a very similar method to what we used to calculate markdowns in Chapter 2.) Some folks find it easier to think about depreciation in this way.

In any case, now that we have our formula, we can find $V(10)$:

$$V(10) = 35000(.88)^{10} = 9747.53 \quad (4.1.16)$$

This means that, after 10 years, the car will be worth \$9,747.53

These two examples just brush the surface of the utility of exponential functions. In the next two sections, we'll investigate how exponential functions are used in financial mathematics.

4.1.3: Exercises

In these exercises, make sure to round all dollar amounts to two decimal places.

1. Find a real-life situation that is modeled using either exponential growth or decay. Explain the situation in 1-2 sentences. If you use a source, please be sure to include a URL or other citation.
2. A population of bacteria in an experiment is modeled by the function $P(t) = 200(1.13)^t$ where t is measured in hours and $P(t)$ is the number of bacteria at time t .
 - a. How many bacteria are there in the population at the start of the study?
 - b. By what percentage does the population increase each hour?
 - c. Find $P(5)$ and round to the nearest whole. Then interpret your answer in the context of the problem, using a complete sentence.
3. You want to save money. You start with \$100 in your piggy bank, and you vow to increase the amount of money in your piggy bank by 10% each week. Write a function that describes the amount of money in your piggy bank after t weeks. How much will you have after one year? (Assume there are 52 weeks in a year.)
4. A restaurant buys an industrial-size freezer. The freezer cost \$10,000 new, and is estimated to have a depreciation rate of 3.1% per year. In 15 years, at what price should the restaurant value the freezer?

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4.2: Simple and Compound Interest

In this section, we'll study how to describe interest accrual using both simple and compound interest, and relate these concepts to our study of exponential functions.

 In this section, you will learn to:

- Use the simple interest formulas to compute and interpret quantities in simple interest scenarios
- Describe how compound interest arises from repeated simple interest calculations
- Use the compound interest formula to compute and interpret quantities in compound interest scenarios

4.2.1: Simple Interest: Ideas and Formulas

The essential idea behind interest is: it costs money to borrow money. There are numerous historical reasons why this is true. You've studied inflation already, so you know that a particular amount of money loses value over time. So, in general, if someone loans someone else money, they expect to get back the amount loaned plus some additional amount — at least enough to make up for inflation, if not more. This additional amount of money owed, beyond the initial amount of the loan itself, is called *interest*. Roughly, *interest* is the fee paid to borrow a certain amount of money.

Interest can be calculated in a variety of ways. You might agree on a fixed extra amount, but what's more common is to describe the interest as percent of the original amount borrowed. As an example: let's say you loan \$3000 to a friend who is trying to start a company, and you ask for 12% interest on your loan. That means that you would expect to be paid back the original \$3000 plus an additional 12% of \$3000 at the end of the time period you mutually specify. (This time period is often called the "term" of the loan.) Using the Basic Percent Equation, we can calculate that 12% of \$3000 is \$360. Therefore, you would expect to get paid back a total of \$3360, \$3000 of which will be repaying your initial investment, and \$360 of which is a fee that you are charging your friend for borrowing your money.

The author acknowledges the many current and historical problems caused by usurious interest rate practices; this book will not cover the very interesting and important debate about whether current interest practices make sense and whether or not interest should exist at all. The fact is that if you want to borrow or loan money through an institutional method (a bank, payday loan company, credit card, or similar structure), there is going to be an interest calculation involved. So we will endeavor to understand some things about these calculations.

This leads to our first definition, which is really an explanation of two formulas:

Definition: Simple Interest

A loan accrues **simple interest**, if the total interest is the product of the initial loaned amount known as the **principal**, a fixed percent known as an **interest rate**, and the length of the loan known as the **term**. The **balance** is the total amount in the account at the end of the term, whose length is a time t in years. We use the formulas:

$$I = Prt \quad \text{and} \quad B = P + I \quad (4.2.1)$$

where I is interest, P is principal, t is time in years, r is the interest rate (as a **decimal**), and B is the balance to describe simple interest calculations.

Notice that we can actually combine these two formulas in the following way by substituting $I = Prt$ into the second formula and then factoring out a P :

$$B = P + Prt = P(1 + rt) \quad (4.2.2)$$

This will give us a quick way to find the balance of a loan -- that is, the total amount that is owed -- if we know the rate, initial amount, and the length of the loan. Let's see an example of how these formulas are used.

✓ Example 4.2.1

Erin takes out a \$1,200 loan to buy a new laptop. The loan has a 4% annual simple interest rate, and a term of 24 months. How much interest will she pay on this loan? How much will she pay in total?

Solution

First of all: if someone knows where to find a loan like this, please contact the author immediately! Unfortunately this is not a realistic situation due to the low interest rate and the simplicity of its structure; rather, it is meant to illustrate the use of the formula using relatively nice numbers. We will see more realistic situations later on.

We need to look through the problem and identify which numbers correspond to which variables in the simple interest formula. Remember that those variables are:

B = Balance
 I = Interest
 P = Principal
 r = rate
 t = time in years

The balance is actually not yet known to us: it's what we're trying to find. This will usually be the case. However, we are told the initial amount of the loan, which is \$1200. The initial value of a loan is called the principal, so we'll use $P = 1200$. The interest rate is 4%, which we'll convert to $r = .04$. In general, the interest rate will be the only percent value in the question. Finally, we're told that the term of the loan is 24 months. We need to convert this to years, and since 24 months is 2 years, we will use $t = 2$. We also don't know the value of I ; we will calculate it from the given information.

Revisiting our variables, we then have:

B = Balance = ?
 I = Interest = ?
 P = Principal = 1200
 r = rate = .04
 t = time in years = 2

The first question asks: how much interest will she pay on this loan? In order to calculate that, we'll use the formula:

$$I = Prt \quad (4.2.3)$$

Substituting in our values $P = 1200$, $r = .04$, $t = 2$ we get:

$$I = 1200 \times .04 \times 2 = 96 \quad (4.2.4)$$

Therefore, she will pay \$96 in interest on this loan.

To find the total amount paid, we can use the formula:

$$B = P + I \quad (4.2.5)$$

Now that we know that $I = 96$, and we previously knew that $P = 1200$, we can calculate:

$$B = 1200 + 96 = 1296 \quad (4.2.6)$$

Therefore, the total amount that she pays back is \$1296.

Now, note that if we'd want to calculate the total amount paid back directly, we could use the more direct formula:

$$B = P(1 + rt) \quad (4.2.7)$$

and calculate:

$$B = 1200(1 + .04(2)) = 1200(1.08) = 1296 \quad (4.2.8)$$

which gives the same answer that we had before.

In general, it's not necessary to calculate the balance twice — the point was to illustrate the utility of the $B = P(1 + rt)$ formula. This will become important in the next section. Before then, though, we illustrate how to find a different value in the simple interest formula.

✓ Example 4.2.2

A bank lends Michael \$20,000 so he can buy a car. Michael signs a contract saying that he'll pay back \$26,000 at the end of 4 years. Assuming this is a simple interest loan, what is its interest rate?

Solution

We'll set up our variables as we did before. We have different information than we did in the previous example. In this case, we know the initial loan amount is \$20,000 so $P = 20000$. Also, we know the final amount paid back — the balance — is \$26000. Therefore, we have $B = 26000$. We're also given that the length of the loan is $t = 4$ years. Looking at our variables, we have:

$$\begin{aligned} B &= \text{Balance} = 26000 \\ I &= \text{Interest} = ? \\ P &= \text{Principal} = 20000 \\ r &= \text{rate} = ? \\ t &= \text{time in years} = 4 \end{aligned}$$

Our goal is to find r , but in order to do that, we need to find I . Otherwise, it will be difficult to find r using the formulas we have. So, our first step will be use the formula $B = P + I$ to solve for I . We have:

$$B = P + I \quad (4.2.9)$$

Substituting in $P = 20000$ and $B = 26000$, we can calculate:

$$26000 = 20000 + I \quad (4.2.10)$$

Now, to find I , we will subtract 20000 from both sides. We get:

$$6000 = I \quad (4.2.11)$$

which means that \$6000 was paid in interest. From there, we can use the formula $I = Prt$ to find r . Substituting $I = 6000$, $P = 20000$ and $t = 4$, we get:

$$6000 = 20000 \times r \times 4 \quad (4.2.12)$$

Now to calculate r , we need to know that the three terms multiplied on the right can have their orders switched. (This is a mathematical property known as *commutativity*.) In any case, we can rewrite the equation above as

$$6000 = 20000 \times 4 \times r \quad (4.2.13)$$

which simplifies to

$$6000 = 80000r \quad (4.2.14)$$

Now, we can use Division undoes Multiplication to get

$$r = \frac{6000}{80000} = .075 \quad (4.2.15)$$

This is the interest rate, expressed as a decimal. So as a percent, the interest rate is 7.5%.

In general, it is possible to manipulate the equations $I = Prt$ and $B = P + I$ to find any of the variables involved, given enough information.

However, it is much more useful in most cases to use the simplified formula $B = P(1 + rt)$ because it leads to the formula for a much more common type of interest: compound interest.

4.2.2: Compound Interest

In the previous section, we studied simple interest. While simple interest is relatively straightforward to compute, it is not the type of interest that is typically used in most actual loans. Instead, most loan accounts use some form of *compound interest*.

Definition: Compound Interest

A loan accrues **compound interest** if the balance is computed as a repeated simple interest calculation on a periodic schedule. The balance on a compound interest loan can be expressed as a function of the time of the loan. This function is given by:

$$B(t) = P \left(1 + \frac{r}{n} \right)^{nt} \quad (4.2.16)$$

where $B(t)$ is the balance at time t , P is the principal, r is the interest rate, n is the number of times per year the interest is calculated, and t is the time in years. This may seem complicated, but we will see soon how it makes sense.

Take a moment to compare the equations for simple vs compound interest. We said that, for simple interest,

$$B = P(1 + rt) \quad (4.2.17)$$

and for compound interest,

$$B(t) = P \left(1 + \frac{r}{n} \right)^{nt} \quad (4.2.18)$$

If you stare at these for a few minutes, you will likely see some similarities. On the left side, there is the balance. (In the second equation, we happened to write the balance as a function of time, but it's still standing for the same thing.) On the right side, there is the principal P , multiplied by a factor that involves the interest rate. However, in the compound interest equation, the variable t is in the exponent. For this reason, compound interest is an *exponential function*.

In the previous definition, we are familiar with all of the variables besides n from the simple interest formulas. The idea behind n is that it counts the number of times per year the interest is calculated. Since the interest rate is annual, we take that rate, r , and divide it by the number of times per year the interest is calculated. This evenly distributes the percent interest calculation throughout the year. However, since the interest is being calculated on a higher and higher balance each time, the amount of interest continues to grow over time.

In general, you can determine what n is by looking for a keyword that indicates the *compounding schedule*. Here is a table that shows the most common values of n and their corresponding keywords. Remember that n is the number of times per year interest is calculated, so these values are obtained by simply counting the number of a given period in a year.

Keyword	n
Annually	1
Monthly	12
Weekly	52
Daily	365

Let's see an example to understand.

Example 4.2.3

Leanne would like to purchase an iPad Pro for \$649 using her credit card. Her credit card has 24% interest rate compounded monthly. Assuming that she does not make any payments on the purchase, how much will she owe after 2 years? Compare this with a simple interest rate for the same rate and time period.

Solution

In order to solve this question, we must note the word "compounded" in the question. That tells us that this is a compound interest question. That tells us to use the formula:

$$B(t) = P\left(1 + \frac{r}{n}\right)^{nt} \quad (4.2.19)$$

where $B(t)$ is the balance at time t , P is the principal of the loan, r is the rate, and n is the number of times per year that the interest is compounded. We are trying to find the balance when the time $t = 2$ years. The initial loan amount, P , is \$649. So we'll plug in $P = 649$. The rate is given by $r = .24$, which is computed by converting the rate of 24% to a decimal.

The only missing variable is n . That can be found by looking for one of the keywords and finding the associated n value. We see in this question that the interest is *compounded monthly*. Therefore, we will choose $n = 12$.

We've shown that:

$$\begin{aligned} B &= \text{Balance} = ? \\ P &= \text{Principal} = 649 \\ n &= \text{monthly} = 12 \\ r &= \text{rate} = .24 \\ t &= \text{time in years} = 2 \end{aligned}$$

From here, we can use the formula: $B(t) = P\left(1 + \frac{r}{n}\right)^{nt}$ with the values given above. That is, we have:

$$B(2) = 649\left(1 + \frac{.24}{12}\right)^{12 \times 2} \quad (4.2.20)$$

Let's make a couple simplifications on the right side:

$$B(2) = 649\left(1 + \frac{.24}{12}\right)^{12 \times 2} = 649(1.02)^{24} \quad (4.2.21)$$

Take a moment to note that this has the same format as the exponential functions we saw in the previous section. We will use the same techniques to calculate it, and we find that:

$$B(2) = 649(1.02)^{24} = 1043.86 \quad (4.2.22)$$

That is, after 2 years, Leanne will owe \$1043.86

To contrast the situation above with simple interest, note that if the same loan were applied with simple interest instead of compound interest, we'd use $r = .24$, $t = 2$, and $P = 649$. That would result in a balance of:

$$B(t) = P(1 + rt) = 649(1 + (.24)(2)) = 960.52 \quad (4.2.23)$$

That is, the same loan with a simple interest structure would be \$960.52 as compared to the \$1043.86 price tag of compound interest. The reason is that compound interest was calculated multiple times over the course of the loan; even though it was a smaller percent interest each time, the exponential growth caused it to grow larger over time.

4.2.3: Exercises

1. You take out a simple interest loan from a bank for \$35,000. The interest rate is 7% and you plan to pay the loan back 3 years later. How much interest will you pay, and how much will you pay in total?
2. An entrepreneur plans to open a coffee shop in Monmouth. A bank finances his equipment by granting him a \$65,000 loan. The loan has 1.7% annual simple interest. He eventually pays the bank \$90,000. How long did it take him to pay back the loan? Give your answer as a number of years rounded to one decimal place.
3. Mr. and Mrs. Gonzales want to save money to help their son Phil go to college. When Phil is born, they deposit \$9,000 into a savings account with 7.3% annual interest that compounds daily. How much will be in the account when Phil turns 18?

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4.3: Annuities

In the previous section, we used simple and compound interest to calculate the value of money invested at a certain interest rate over a period of time. In these situations, the money was only transferred twice: once at the start of the loan from the lender to the borrower, and once at the end of the loan in the opposite direction. For some situations, this kind of borrowing or investing makes sense. However, for many people who are trying to save up for a large expense, such as a car, house, or retirement fund, it's not reasonable to invest a large lump sum in an account at its start, and simply wait for interest to accrue.

For example, let's say you wanted to save up for your retirement. You plan to retire in 40 years, and you'd like to save \$1,000,000 by that point. Even with an excellent savings account rate of 5%, and with interest compounded monthly, you'd need to initially deposit \$135,898.79 in the account. This simply isn't an amount of money that most people have lying around to put into an account!

4.3.1: Ordinary Annuities

Fortunately, there is another savings structure that allows someone to deposit smaller amounts of money regularly over time to save for a large savings goal. This structure is known as an *ordinary annuity*.

Definition: Ordinary Annuity

An **ordinary annuity** is an account into which a sequence of equal, regular payments are made, and that receives compound interest on those deposits. The deposits are made with the same regularity that the interest compounds. The formula that describes annuities is:

$$B(t) = \left(\frac{n \cdot pymt}{r} \right) \cdot \left(\left(1 + \frac{r}{n} \right)^{(nt)} - 1 \right) \quad (4.3.1)$$

where $B(t)$ is the balance at time t , r is the annual interest rate, n is the number of times per year interest compounds, $pymt$ is the periodic payment, and t is time in years.

Usually annuities have an agreed-upon time frame (meaning that a person agrees to pay the same amount regularly over some number of years). Once the time has elapsed, the annuity **matures**. The balance at that time is called the **future value** of the annuity.

Looking at the definition above, we see some familiar elements. The variables $B(t)$, r , n , and t are all familiar to us from our study of compound interest, and stand for exactly the same things. The only new variable here is $pymt$, which stands for the periodic payment. In practice, this is usually a monthly payment, but can be paid at any schedule (quarterly, daily, etc.). The timing of the payment and the interest compounding are assumed to be the same in this text (though, in some real-world cases, they are not).

We also notice that part of the annuity equation is the compound interest equation! That is due to the fact that annuities are built on compound interest repeated many times. That is, each time a deposit is made, the compound interest is recalculated on the increased amount in the account. At the start of an annuity, the amount in the account is quite small. But over time, the account grows rapidly for two reasons. First, money is being deposited into the account at regular intervals. But also, interest is being compounded on those new deposits! Annuities are a great way to save money if you don't have a lot at the start, but can reliably make deposits into an account over a long period of time. Most Independent Retirement Accounts (IRAs) have the form of an ordinary annuity, potentially with some slight modifications.

Let's see an example that illustrate how ordinary annuities work, and why they are often more useful for people than savings accounts in saving for large expenditures.

✓ Example 4.3.1

A certain math professor would like to retire in 10 years. (Just kidding. There is no way that is going to happen.) She opens an ordinary annuity with 3.6% annual interest compounded monthly. She makes monthly payments of \$200 into the annuity. Write the function describing the balance in the annuity at time t . How much will she have when the annuity matures?

Solution

This question uses an ordinary annuity, which means we need to identify the variables in the ordinary annuity formula before we can proceed. We see that the length of the annuity is 10 years, so we'll use $t = 10$. The interest rate is 3.6%, so we'll use $r = .036$. Since the deposits are made monthly, that means that interest also compounds monthly, since the deposits and the compounding are made on the same schedule. Therefore, we'll use $n = 12$. Finally, the monthly payment is \$200, so we'll use $pymt = 200$.

We can organize this information below, which is recommended whenever you are solving an annuity problem:

$$\begin{aligned} B &= \text{Balance} = ? \\ pymt &= \text{Payment} = 200 \\ n &= \text{monthly} = 12 \\ r &= \text{rate} = .036 \\ t &= \text{time in years} = 10 \end{aligned}$$

The first question asks us to find a function that describes the balance of the account after t years. To do this, we'll use the ordinary annuity formula, with all of the variables except for t plugged in. We have:

$$B(t) = \left(\frac{12 \cdot 200}{.036} \right) \cdot \left(\left(1 + \frac{.036}{12} \right)^{(12t)} - 1 \right) \quad (4.3.2)$$

Take a moment to note that the equation above actually gives us a lot of information: we could plug in any value of t and find the balance of the account after that number of years. However, we are interested in what happens when $t = 10$ years. So, we'll plug in $t = 10$. We have:

$$B(10) = \left(\frac{12 \cdot 200}{.036} \right) \cdot \left(\left(1 + \frac{.036}{12} \right)^{(12 \times 10)} - 1 \right) \quad (4.3.3)$$

The value on the right side of the equation above is the answer. However, we need to calculate it to get a meaningful answer that corresponds to a dollar amount. There are numerous ways to do this. For the easiest way, see [this video](#). We'll show a slightly more complicated way here that will lead to the same final answer, just for reference.

First, we'll simplify the first set of parentheses, perform the division and addition in the second set of parentheses, and the multiplication in the exponent. Since we are midway through the problem, we'll write out the entire decimal expansion visible in the calculator window to get maximum accuracy in the final answer. We have:

$$B(10) = 66666.66667 \cdot \left((1.003)^{120} - 1 \right) \quad (4.3.4)$$

Next, we'll calculate the expression $(1.003)^{120} - 1$. We find:

$$B(10) = 66666.66667 \cdot .4235571693 \quad (4.3.5)$$

Finally, we'll multiply these two values together. We get:

$$B(10) = 28837.14462 \quad (4.3.6)$$

That is, the balance after 10 years is \$28,837.14

There are several things to note about the example above. First of all: \$28,837.14 is not enough money to have saved up to retire comfortable for the vast majority of people, so sadly, this math professor won't be able to retire in 10 years with this sort of savings plan. However, it does show how money can accumulate to a relatively large amount, even though the individual deposits are relatively small.

Moreover, the annuity described above is better than simply putting \$200 in a shoebox each month for 10 years. Simply putting away \$200 per month results in setting aside the same amount of money, but without the compound interest built in. If you simply set aside \$200 each month without investing it, you'd be making 120 payments of \$200 each, which would result in \$24,000 after

10 years. This is less than the \$28,837.14 in the annuity calculation. The difference is due to the compound interest that is calculated on each deposit.

The question above illustrates how ordinary annuities work in general. But what if you wanted to save up for a particular amount at a particular time, and you wanted to know how much to deposit each month to get there? That situation arises quite often -- any time that you are trying to save for a particular goal, you would ask yourself this question. Let's see an example to illustrate how this works.

✓ Example 4.3.2

Find the size of the monthly payments necessary to accumulate \$1 million at 5% annual interest compounded monthly on an ordinary annuity over 40 years.

Solution

This problem refers to a situation earlier in this section -- you'd like to save \$1,000,000 in 40 years. How do you get there? In this case, we'll use the ordinary annuity formula to find the value of $pymt$, which will represent the monthly payment in this case.

The other variables in the ordinary annuity formula can also be identified in this question. We have that deposits are made monthly, so we'll use $n = 12$. We are interested in the value after 40 years, so we'll use $t = 40$. The interest rate, as before, is 5%, so we'll use $r = .05$. Finally, we would like the balance to be \$1,000,000. This means that $B(40) = 1000000$.

Now, instead of finding the function as we did in the previous problem, we'll go ahead and plug in our values for $B(t)$, r , n , and t . And then we'll solve for $pymt$. Using the ordinary annuity formula with our variables plugged in, we have:

$$1000000 = \left(\frac{12 \cdot pymt}{.05} \right) \cdot \left(\left(1 + \frac{.05}{12} \right)^{(12 \times 40)} - 1 \right) \quad (4.3.7)$$

Let's start by simplifying on the right first. We can calculate the second set of parentheses and reduce the equation to:

$$1000000 = \left(\frac{12 \cdot pymt}{.05} \right) \cdot (6.358417319) \quad (4.3.8)$$

Next, we'll divide both sides by the 6.358417319 value. We have:

$$\frac{1000000}{6.358417319} = \frac{12 \cdot pymt}{.05} \quad (4.3.9)$$

Next, we'll use Cross Multiplication to eliminate the fractions. We have:

$$76.30100782 \times pymt = 50000 \quad (4.3.10)$$

where the 76.30100782 value is obtained by multiplying 12×6.358417319

Finally, we'll use Division undoes Multiplication to find the value of $pymt$:

$$pymt = \frac{50000}{76.30100782} = 655.2993391 \quad (4.3.11)$$

That is, we should be making monthly payments of \$655.30 to reach \$1,000,000 after 40 years.

Spending \$655 per month on a savings goal is a lot of money! However, for most people, that is still a more attainable goal to reach than an initial deposit of nearly \$136,000 in a savings account to reach \$1,000,000. Additionally, we can see the power of compound interest magnified here: if we simply deposited \$655.30 into a shoebox every month for 40 years, we'd only end up with \$314,544. That means that *almost \$700,000 of additional money is generated through compound interest*. This is why many people use retirement accounts such as IRAs instead of simply setting aside money in a non-interest-bearing account.

We've seen how to save up for a particular goal. What happens if we have a large amount of money, and would like to receive regular outflows from it, while the remaining amount in the account continues to accrue interest? We'll see how that works next.

4.3.2: Payout Annuities

Let's say you have a large amount of money — say, the amount of money you've put in a retirement savings account — and you'd like to take out small amounts of money over time, while the remaining money in the account accrues compound interest. This is a structure known as a *payout annuity*, and it is the basic structure behind how most pension retirement accounts function. Here's the basic definition, which has a rather complicated formula.

Definition: Payout Annuity

A **payout annuity** is an account that pays out equal payments each period over a specified amount of time, while the remaining balance in the annuity continues to earn compound interest. The formula that describes payout annuities is:

$$P = \left(\frac{n \cdot pymt}{r} \right) \cdot \left(\frac{\left(\left(1 + \frac{r}{n} \right)^{nt} - 1 \right)}{\left(1 + \frac{r}{n} \right)^{nt}} \right) \quad (4.3.12)$$

where P is the principal amount invested in the account, r is the annual interest rate, n is the number of payouts each year, $pymt$ is the amount paid out each period, and t is time in years.

Now, the right side of the equation above is quite intimidating. However, you can see some familiar parts cropping up -- the compound interest equation is there, as well as parts of the ordinary annuity formula. Basically, this formula compares the compound interest formula to the ordinary annuity formula, and calculates the difference. That is, we can use it to find the principal amount invested given a desired periodic payout.

Let's see a final example that shows how to use this formula.

✓ Example 4.3.3

Marissa has a payout annuity that pays out \$2,000 each quarter. If the money is invested at 7.2% annual interest, compounded quarterly, and is intended to continue for 15 years, how much was originally deposited?

Solution

This is a payout annuity question, so we'll start by identifying the variables in the payout annuity formula. We have that $pymt = 2000$ since there is a $\$2000$ payout each quarter. Since the payouts are quarterly, we have $n = 4$. We have that the interest rate is 7.2%, and thus we'll use $r = .072$. Finally, we have that $t = 15$, since this annuity lasts 15 years. Therefore, using the payout annuity formula, we have:

$$P = \left(\frac{4 \cdot 2000}{.072} \right) \cdot \left(\frac{\left(\left(1 + \frac{.072}{4} \right)^{(4 \times 15)} - 1 \right)}{\left(1 + \frac{.072}{4} \right)^{(4 \times 15)}} \right) \quad (4.3.13)$$

As before, you can enter this into Wolfram Alpha as a complicated expression, but you may also want to know how to calculate this in another way. Indeed, it's not so bad. First, we'll simplify the expressions that look like $\left(1 + \frac{.072}{4} \right)^{(4 \times 15)}$. We have

$$P = \left(\frac{4 \cdot 2000}{.072} \right) \cdot \left(\frac{\left((1.018)^{60} - 1 \right)}{(1.018)^{60}} \right) \quad (4.3.14)$$

Next we'll calculate the expressions of the form $(1.018)^{60}$ and the first set of parentheses. We get that

$$P = (111111.1111) \cdot \frac{(2.916531558 - 1)}{2.916531558} \quad (4.3.15)$$

Next, we can subtract on the top of the fraction:

$$P = (111111.1111) \cdot \frac{1.916531558}{2.916531558} \quad (4.3.16)$$

Next, we'll simplify the fraction by dividing:

$$P = (111111.1111) \cdot (.6571269743) \quad (4.3.17)$$

Finally, multiplying, we get that:

$$P = 73014.10825 \quad (4.3.18)$$

That is, Marissa will need to initially deposit \$73,014.11 in the account to have the desired payout.

While this is a large amount of money, the result of having such an investment is a payout of \$2000 each quarter for 15 years. That would be a total of \$120,000 that is paid out. So, on an initial investment of \$73,014.11 that is a very good deal!

4.3.3: Exercises

Almost all of the answers on this homework are dollar values. In the middle of a problem, if you calculate a decimal, please do not round. However, at the end of the problem, please do round to two decimal places (the cents place).

1. Linda is 25 when she opens a retirement savings annuity. She deposits \$100 each month into the account, which has 2.4% annual interest compounded monthly.
 - a. Write a function $B(t)$ that describes the balance in Linda's account t years after she opens it.
 - b. Using the function above, determine the amount of money in the account when Linda is 65.
2. Kai needs to have \$800,000 saved by the time that they retire. They plan to retire in 45 years, and they find an ordinary annuity with 1.2% annual interest compounded monthly. What size monthly deposits will they need to make to reach their goal?
3. When Sarah retires, she wishes to make a retirement income of \$2500 per month. She has found a payout annuity that has 4.8% annual interest compounded monthly. She'd like to have this income for 30 years after she retires. What amount will she need to initially deposit in the payout annuity to make this guaranteed retirement income?
4. Jethro intends to retire in 15 years. He can afford to put \$250 per month into an account that accrues 1.8% annual interest compounded monthly to save for his retirement.
 - a. Is this situation better described by an ordinary annuity or a payout annuity? Explain in one sentence.
 - b. Find out how much Jethro will have if he saves \$250 per month using this account. Show your work.
5. Jaime has just retired and has managed to save \$1,200,000 for retirement. He plans to live on this money for 40 years. He has the money in an account out of which he plans to make regular monthly withdrawals while the remaining balance accrues 3.6% annual interest compounded monthly.
 - a. Is this situation better described by an ordinary annuity or a payout annuity? Explain in one sentence.
 - b. Find out how much Jaime will be able to withdraw monthly using this account. Show your work.

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